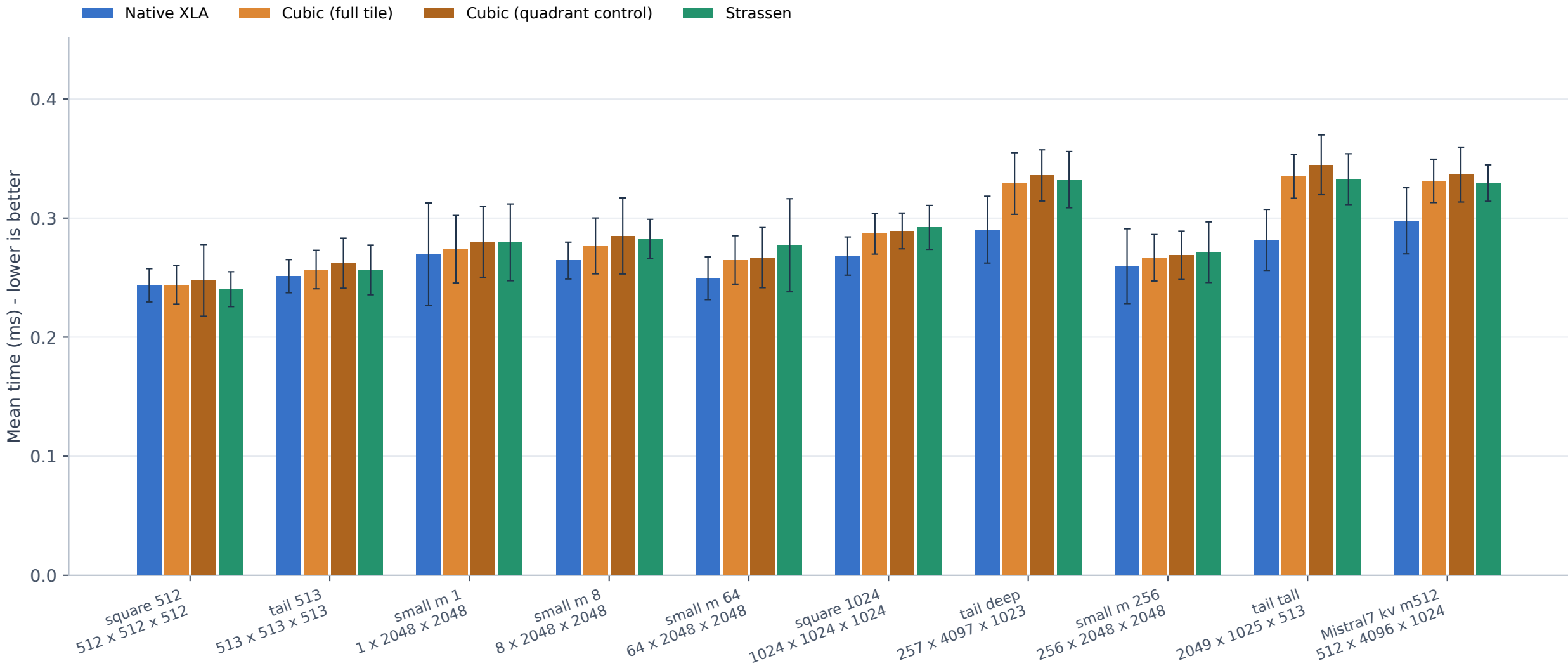


N1 | Fixed-tile baselines - shapes 1-10

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.

Complete call | Original N1-N4 allocation

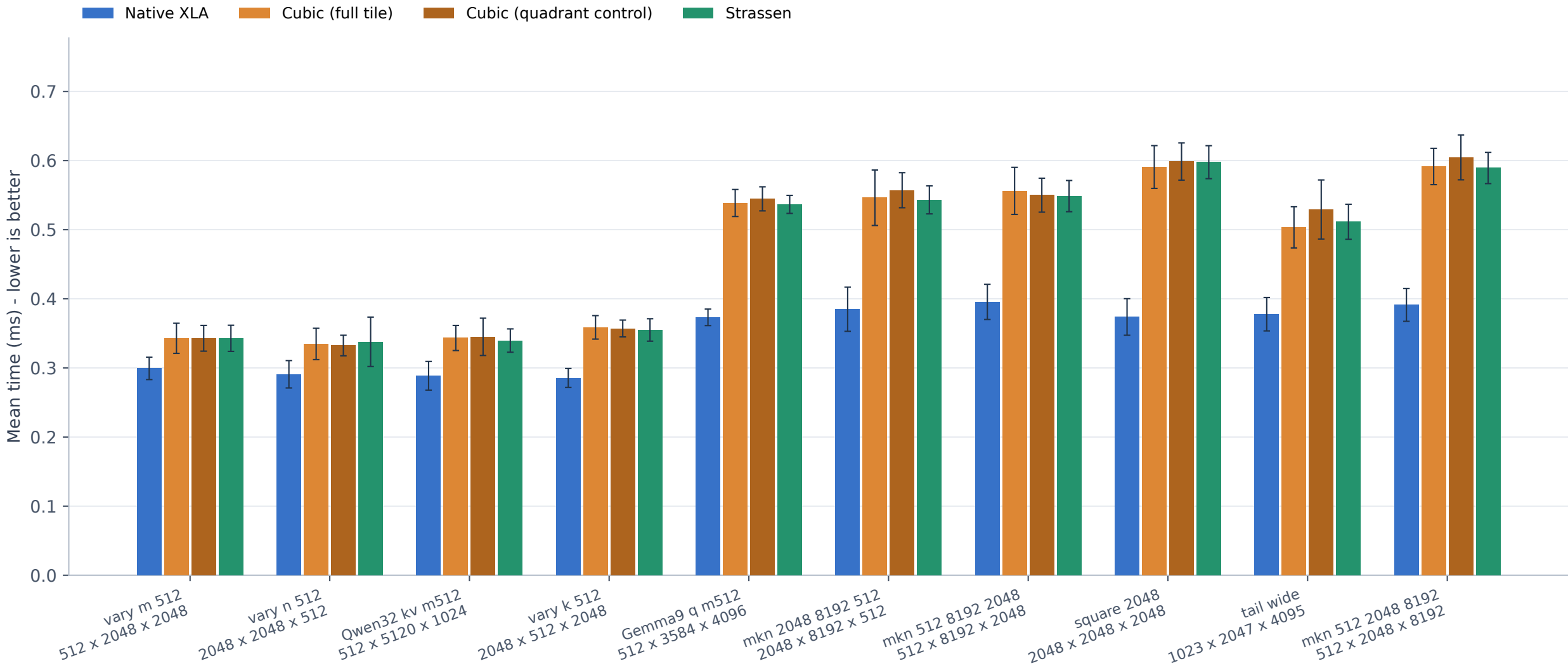


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N1 | Fixed-tile baselines - shapes 11-20

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.

Complete call | Original N1-N4 allocation

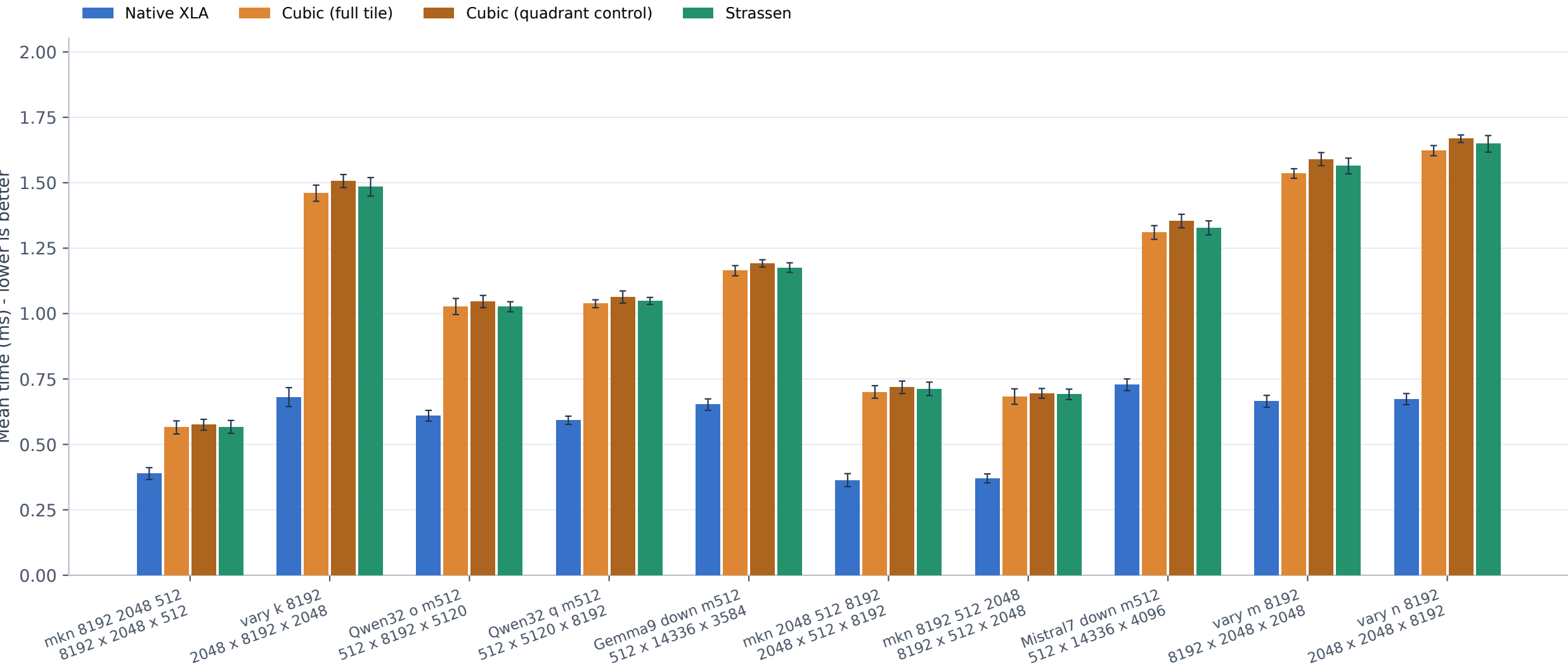


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N1 | Fixed-tile baselines - shapes 21-30

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.

Complete call | Original N1-N4 allocation

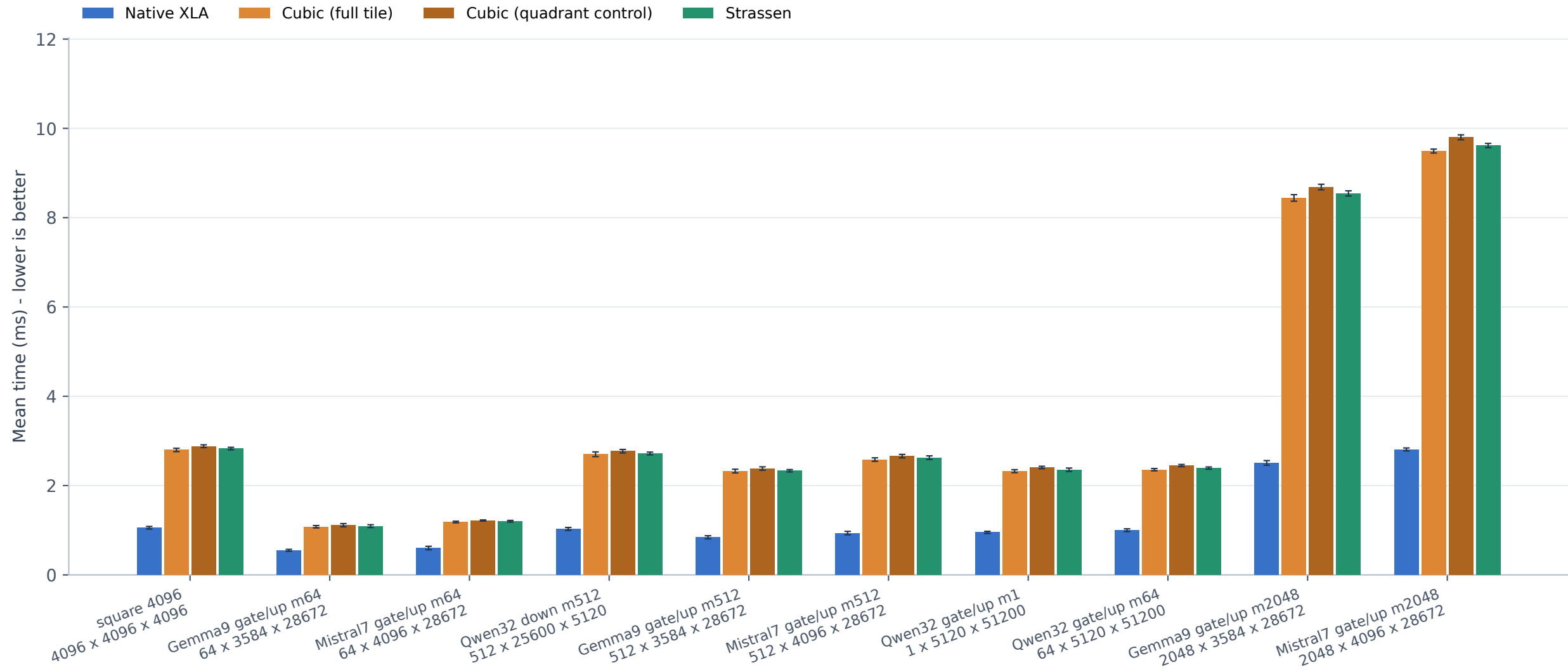


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N1 | Fixed-tile baselines - shapes 31-40

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.

Complete call | Original N1-N4 allocation

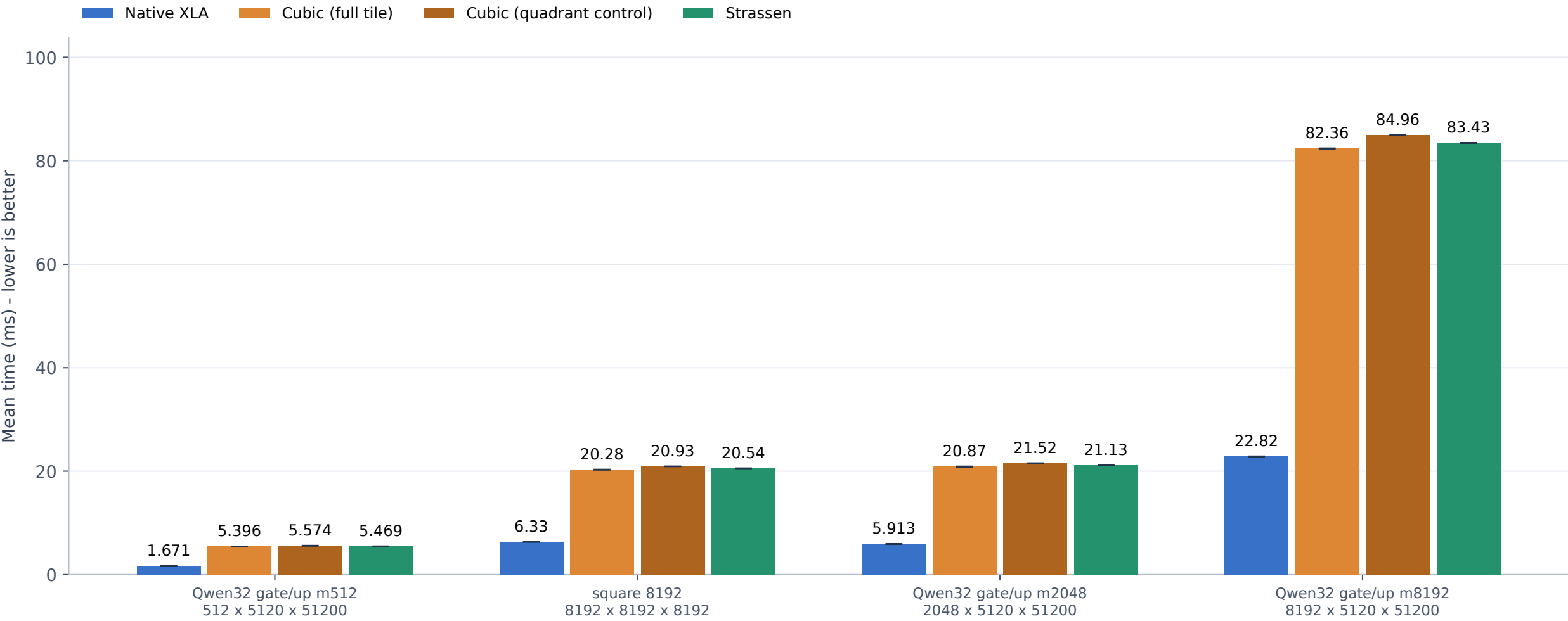


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N1 | Fixed-tile baselines - shapes 41-44

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.

Complete call | Original N1-N4 allocation

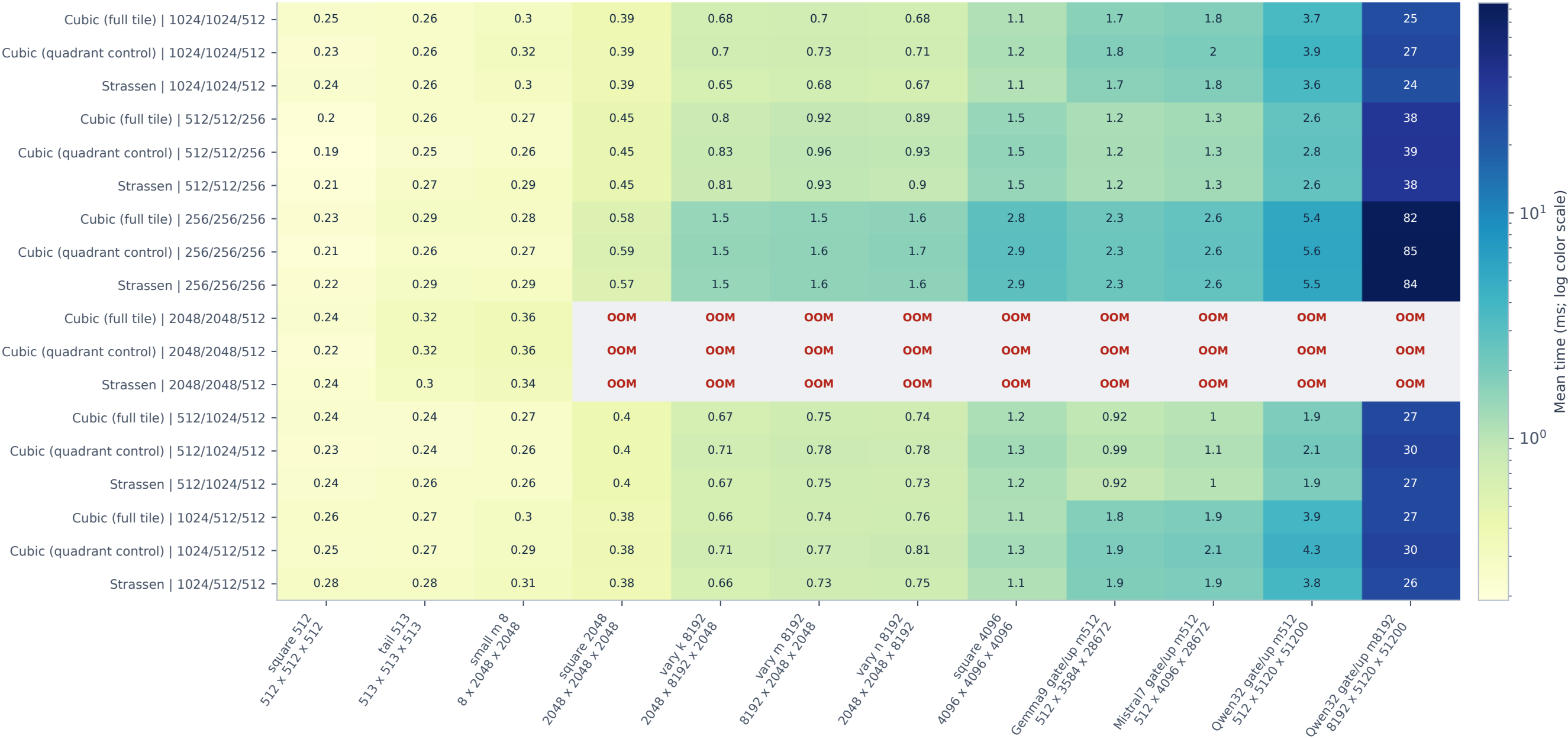


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N2 | Every tile and algorithm

Exploratory screen: 7 timings per passing cell. Tile order is BM/BN/BK. Native was not measured.

Complete call | Original N1-N4 allocation

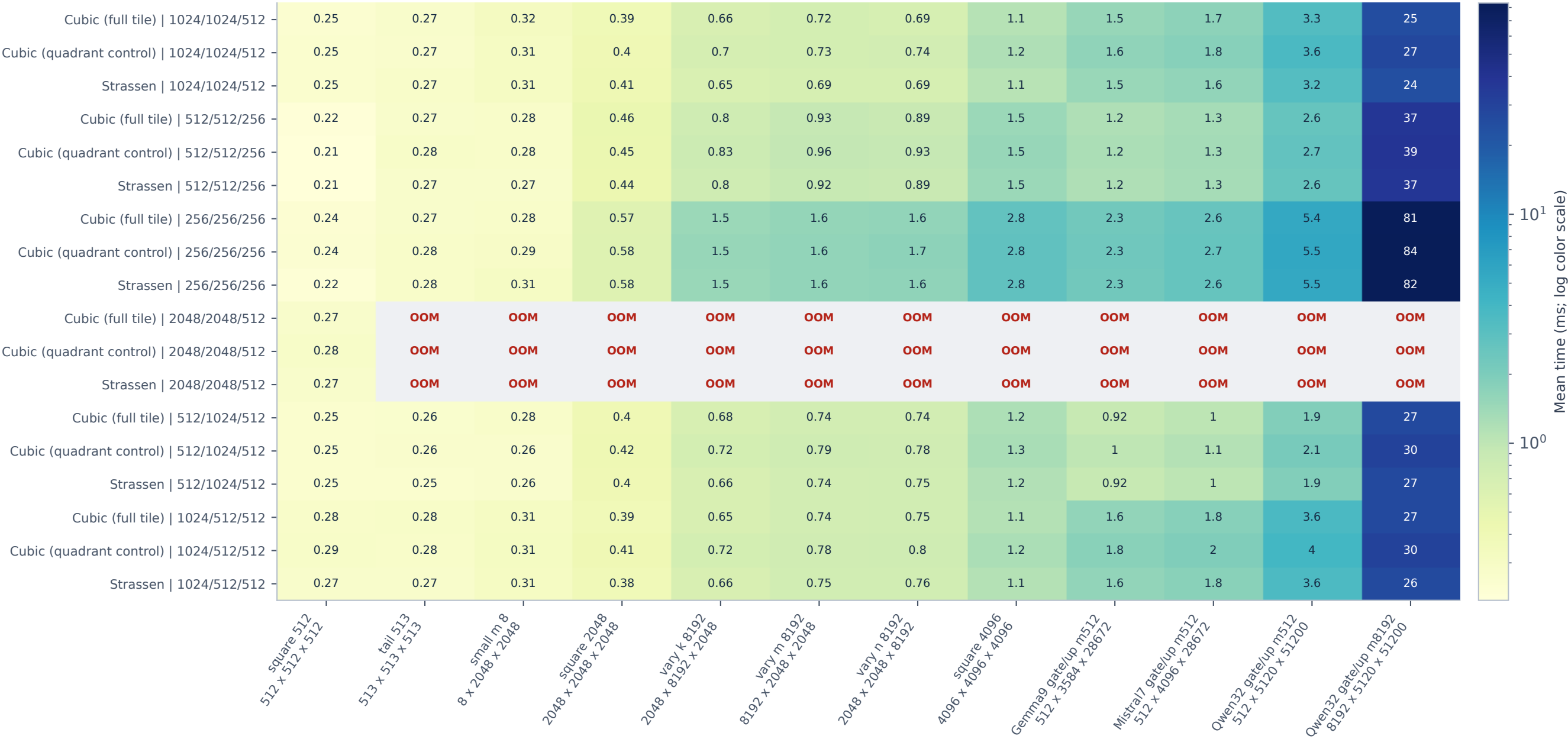


Numbers are archived measurements; missing cells are not measured or failed.

N2 | Every tile and algorithm

Exploratory screen: 7 timings per passing cell. Tile order is BM/BN/BK. Native was not measured.

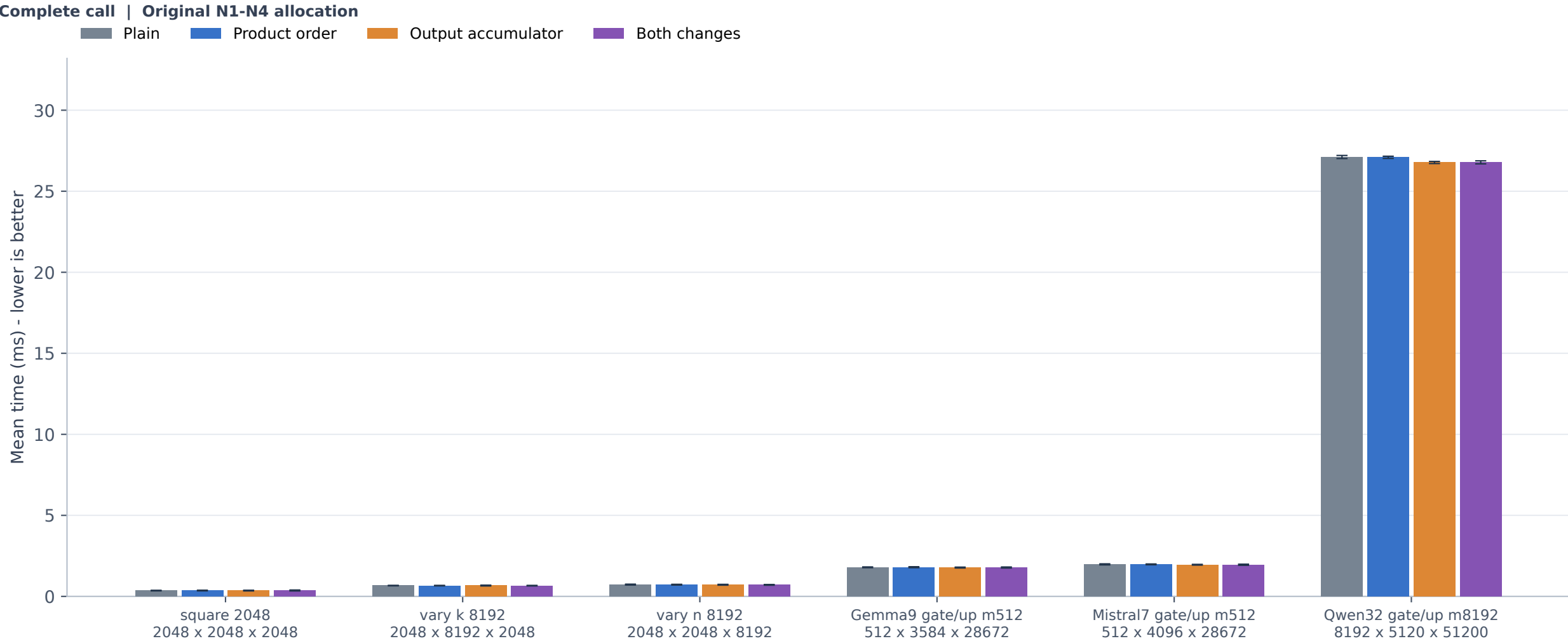
Prepared kernel | Original N1-N4 allocation



Numbers are archived measurements; missing cells are not measured or failed.

N3 | Cubic (quadrant control) optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



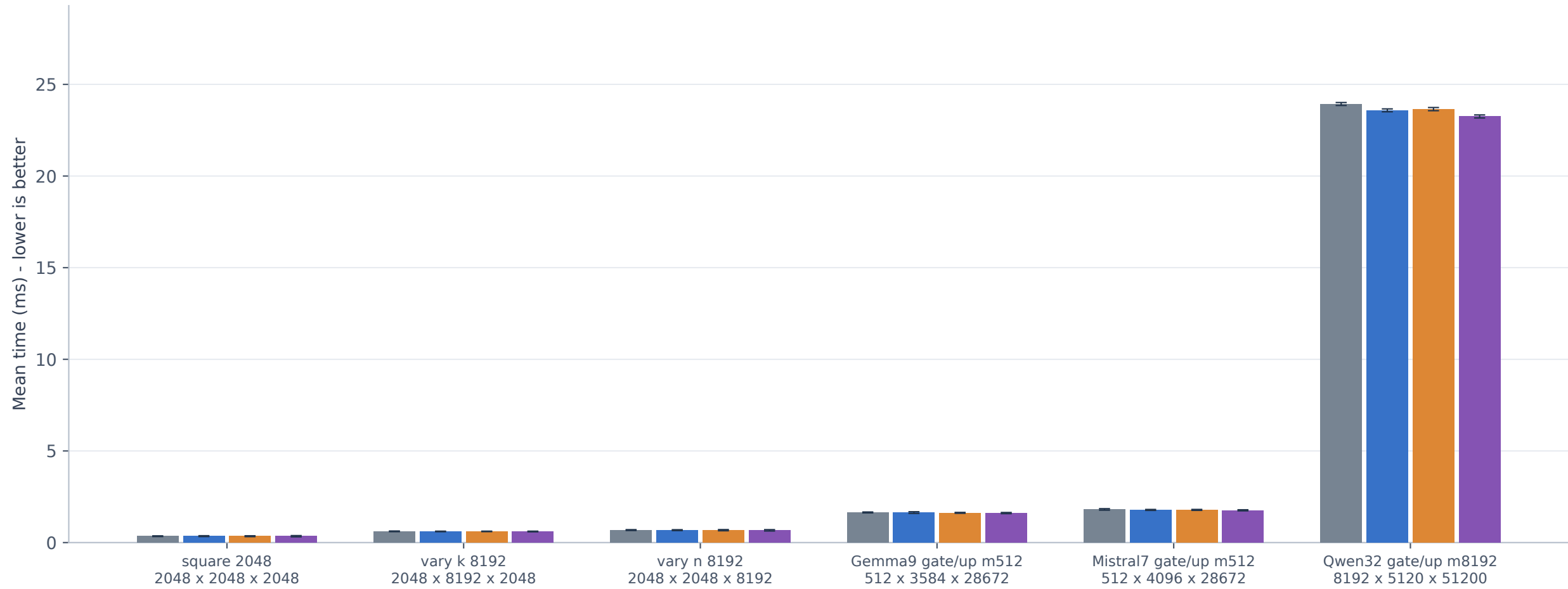
Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N3 | Strassen optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.

Complete call | Original N1-N4 allocation

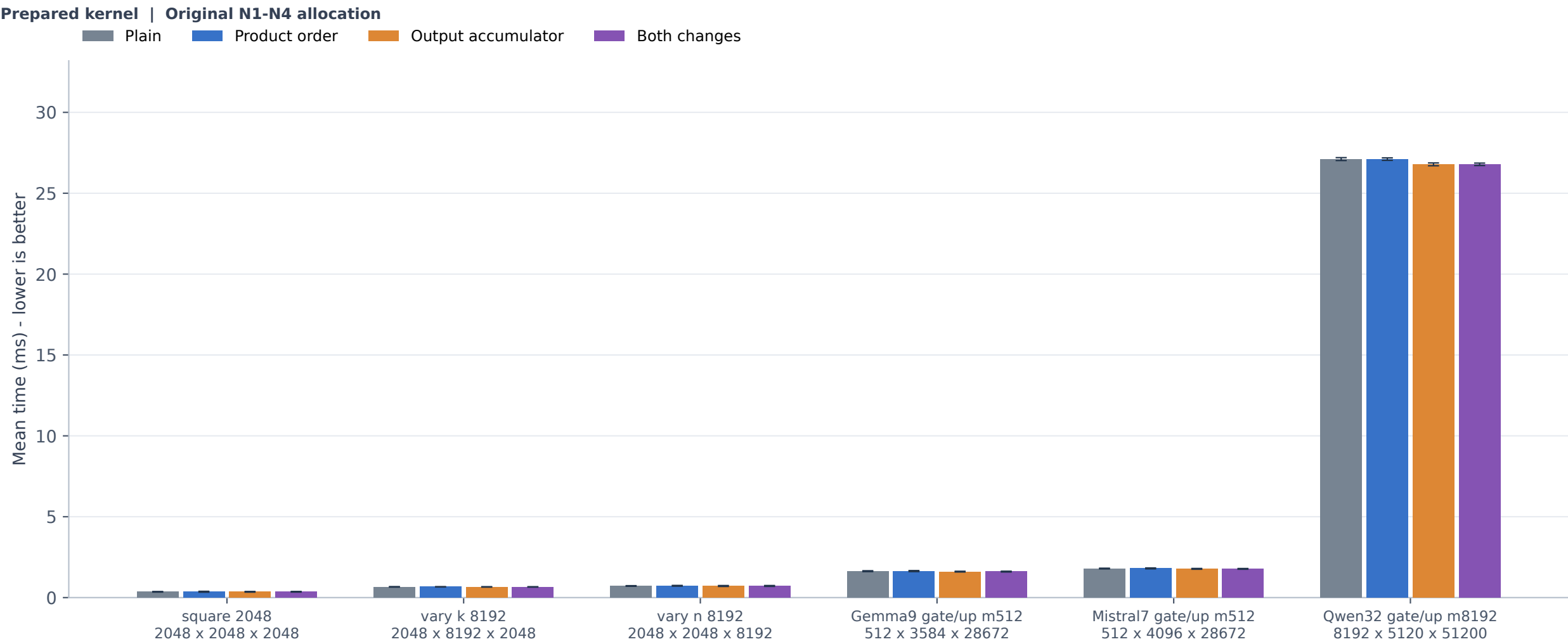
Plain Product order Output accumulator Both changes



Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N3 | Cubic (quadrant control) optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



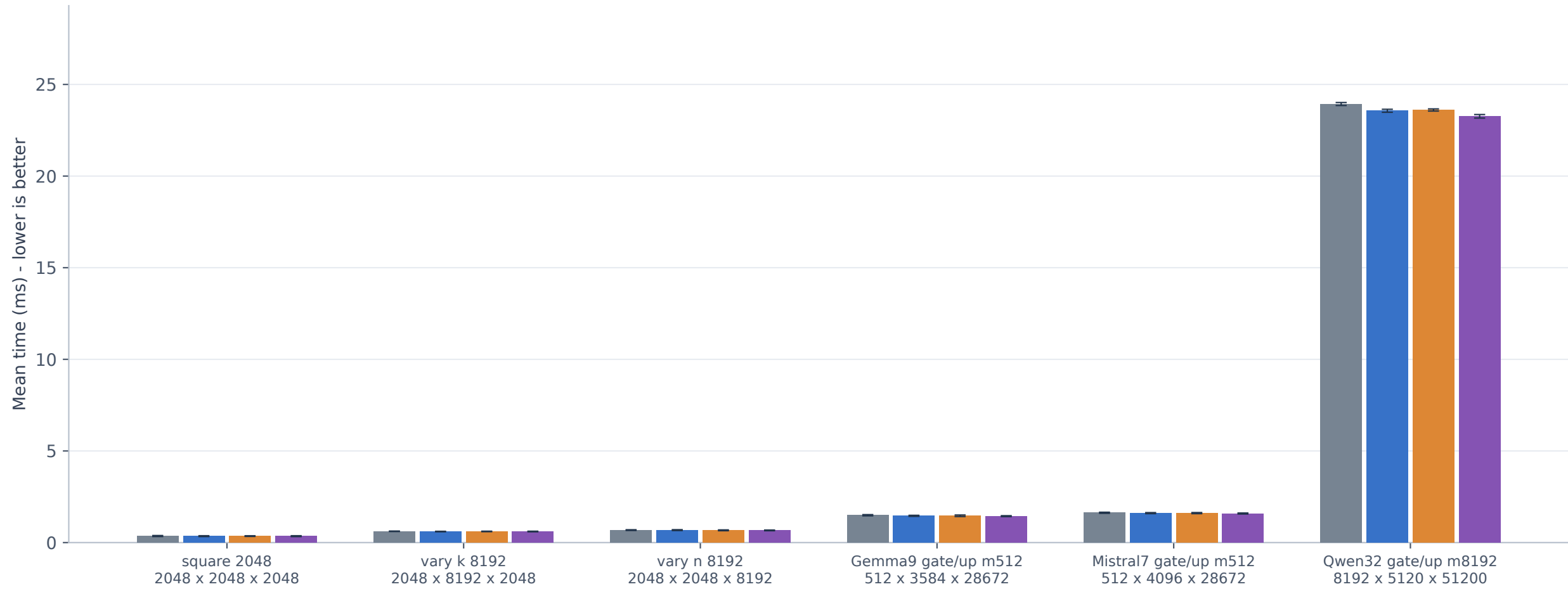
Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N3 | Strassen optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.

Prepared kernel | Original N1-N4 allocation

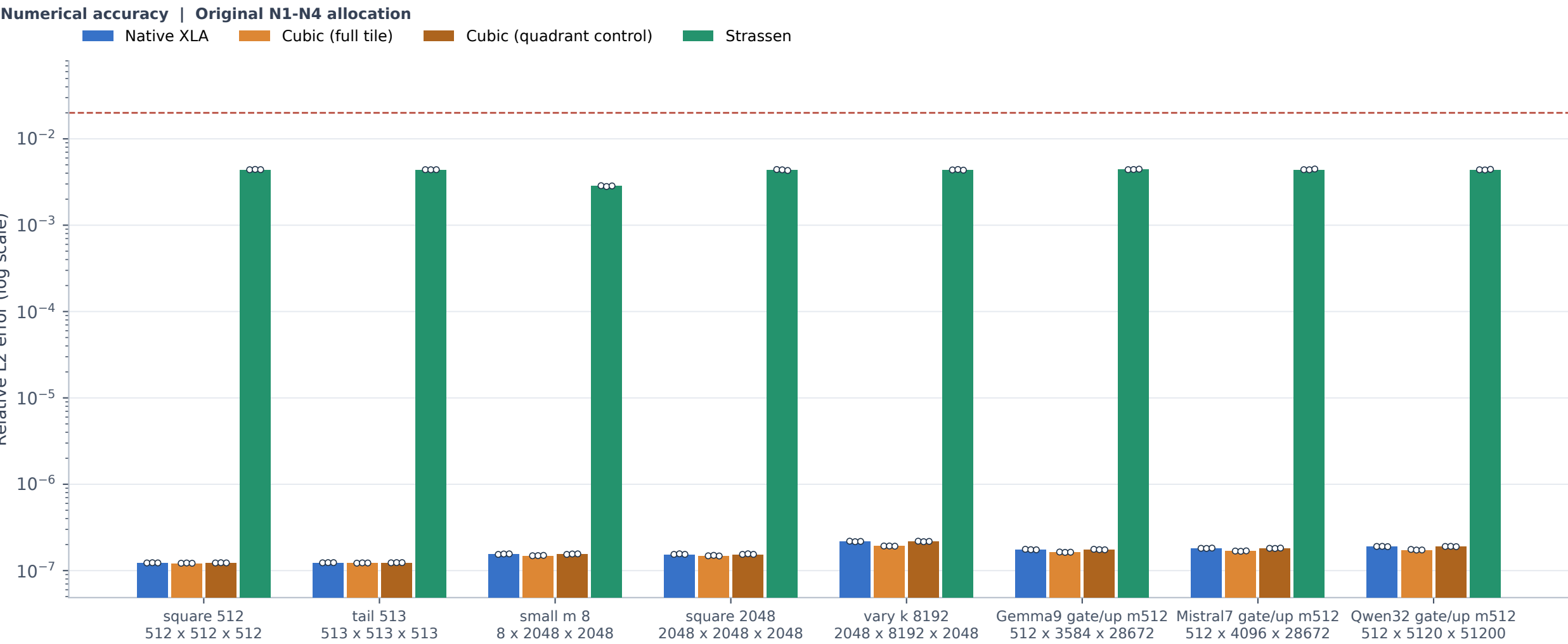
Plain Product order Output accumulator Both changes



Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N4 | gaussian - relative error

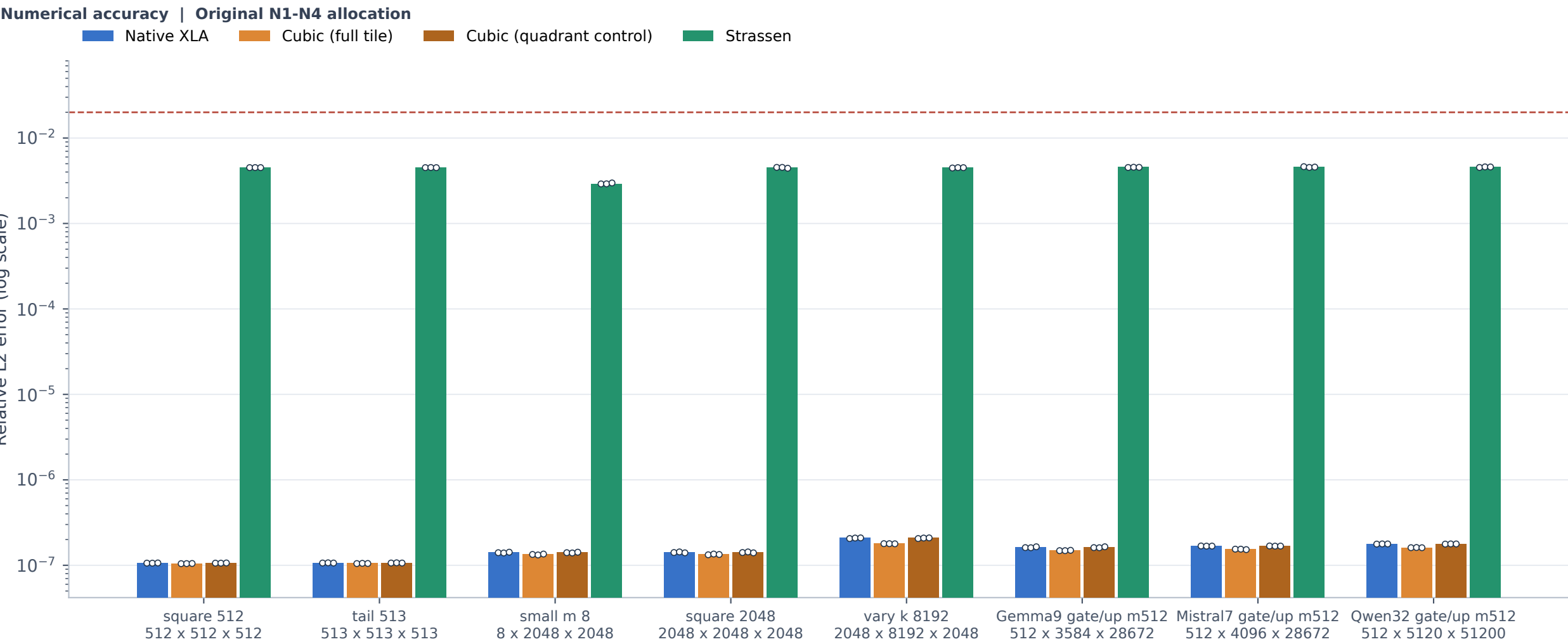
Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N4 | uniform - relative error

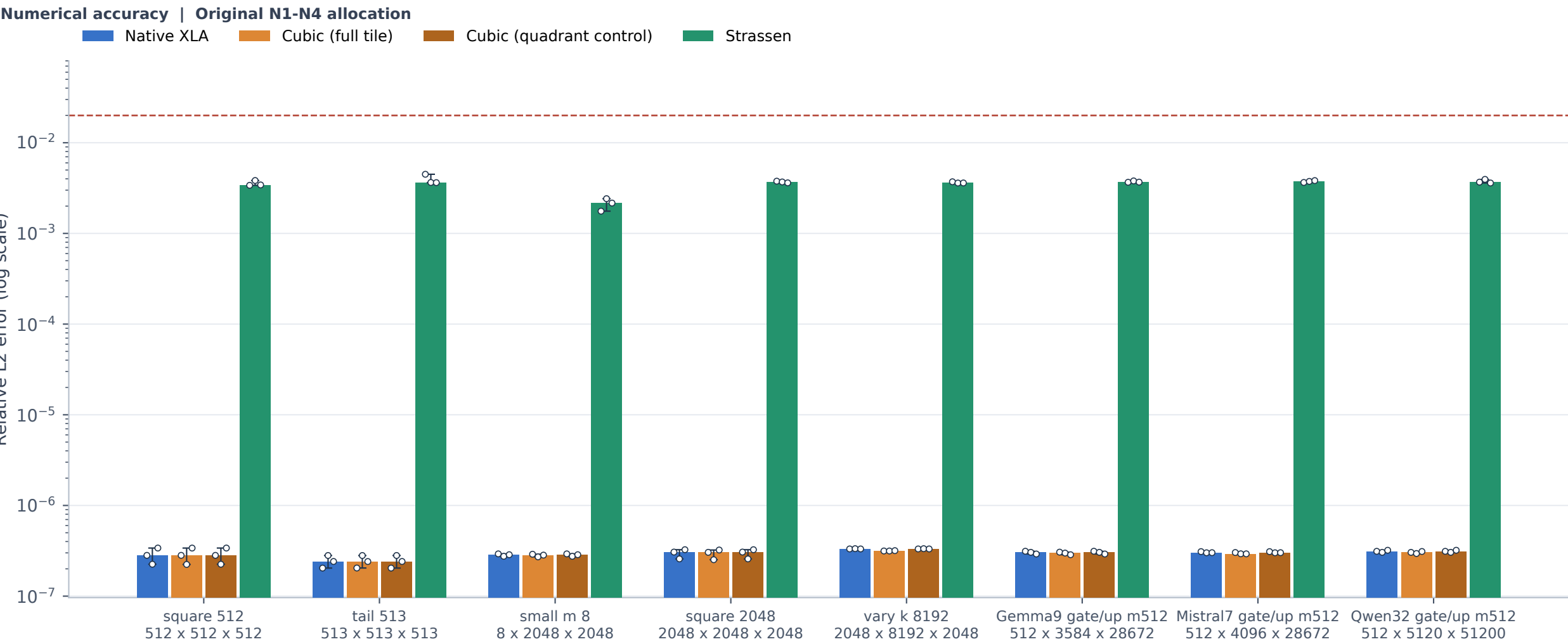
Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N4 | log scale - relative error

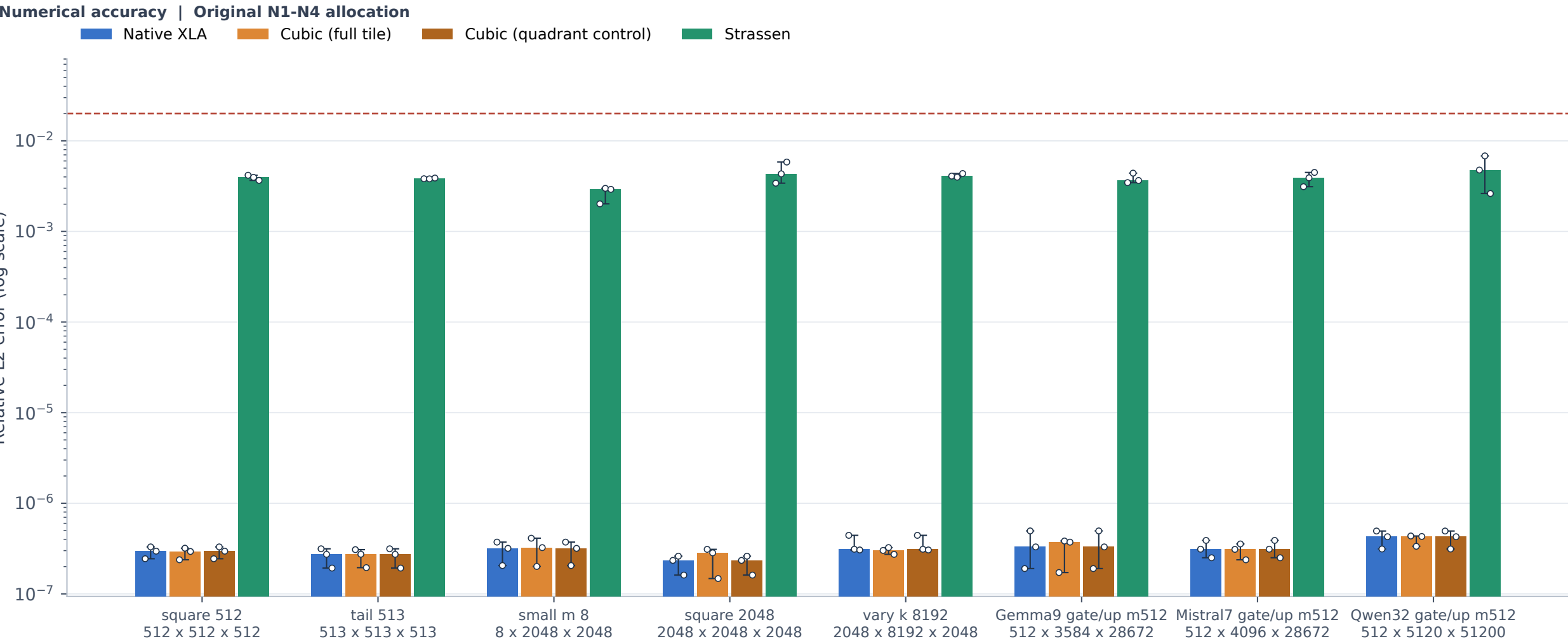
Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N4 | outliers - relative error

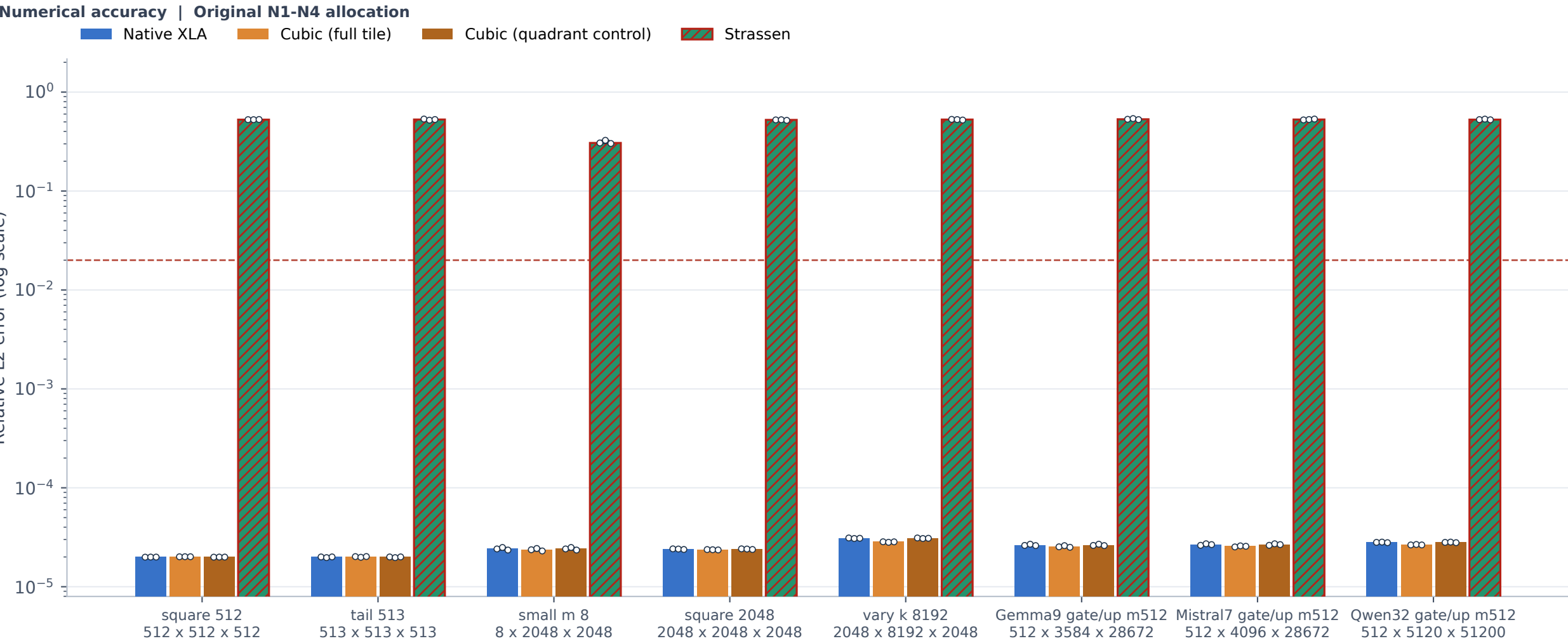
Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N4 | cancellation - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.

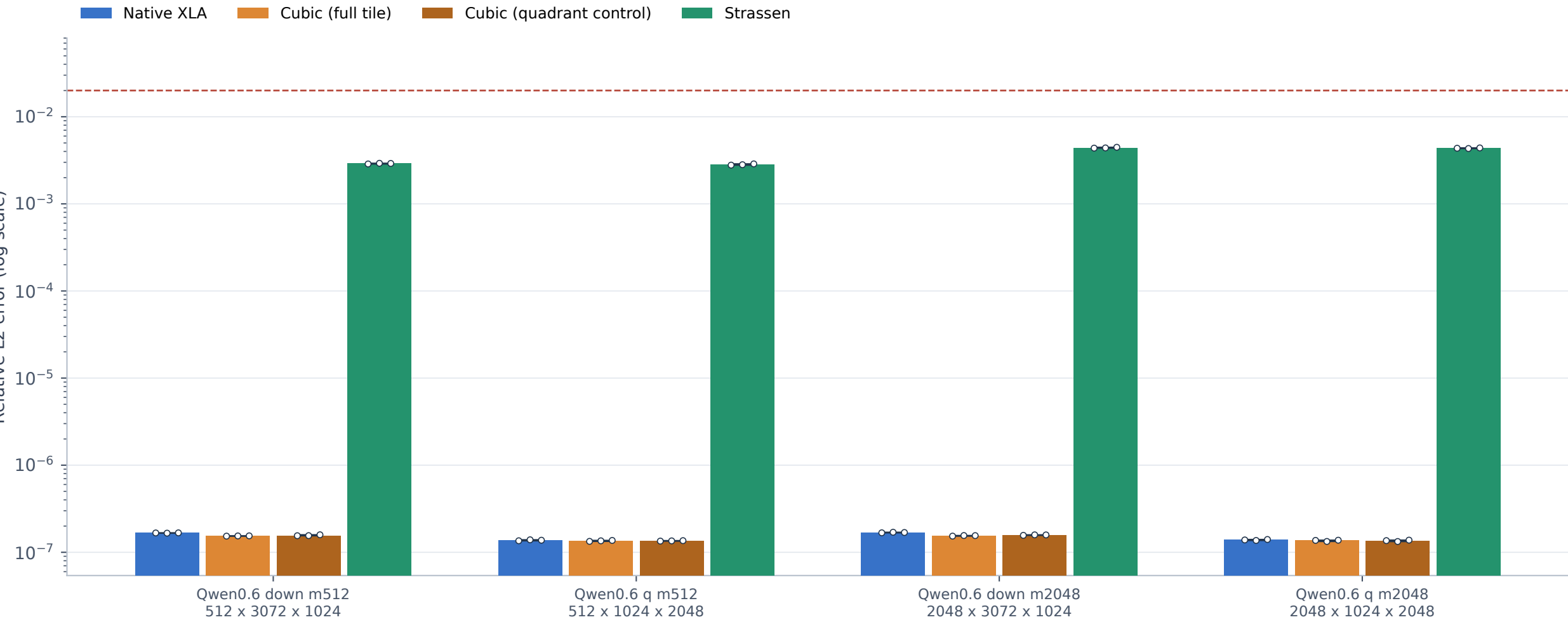


Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N4 | Actual Qwen weights - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.

Numerical accuracy | Original N1-N4 allocation

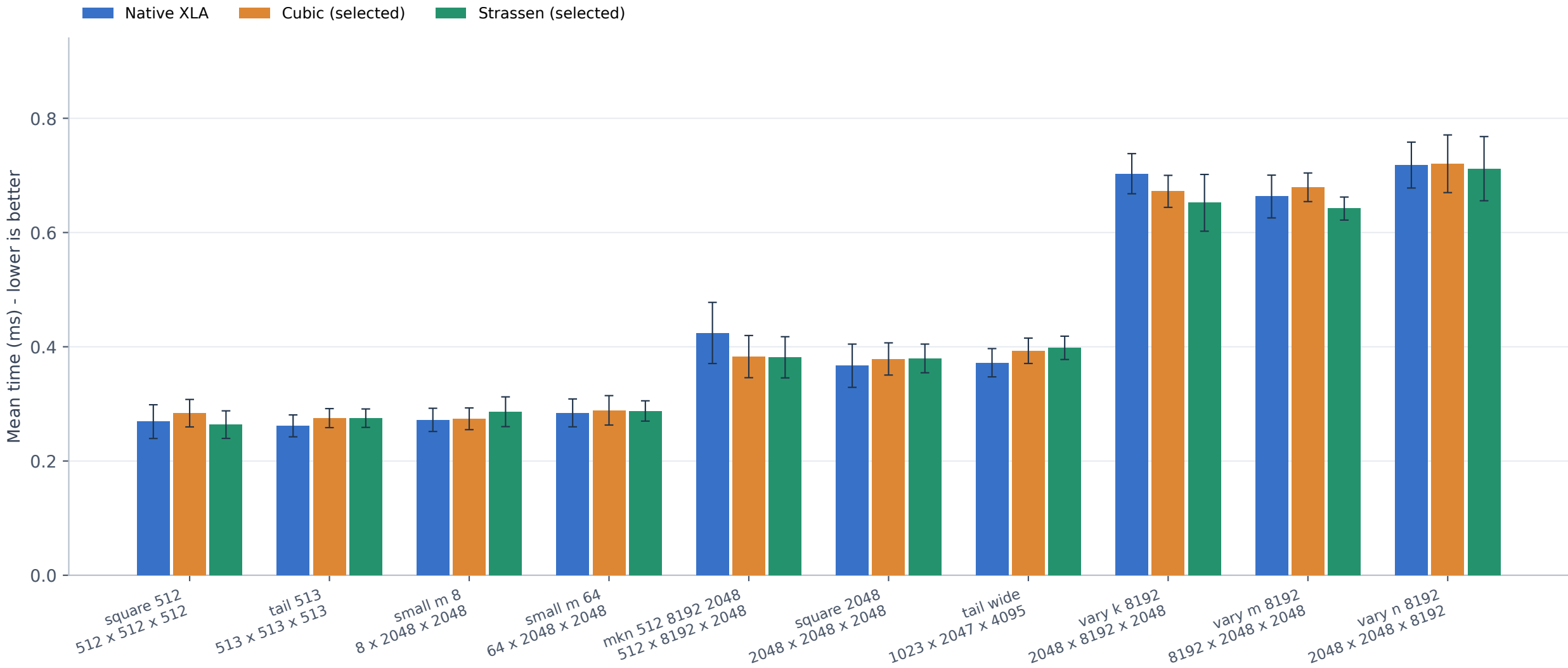


Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

N5 | Fresh confirmation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | N5-N9 allocation

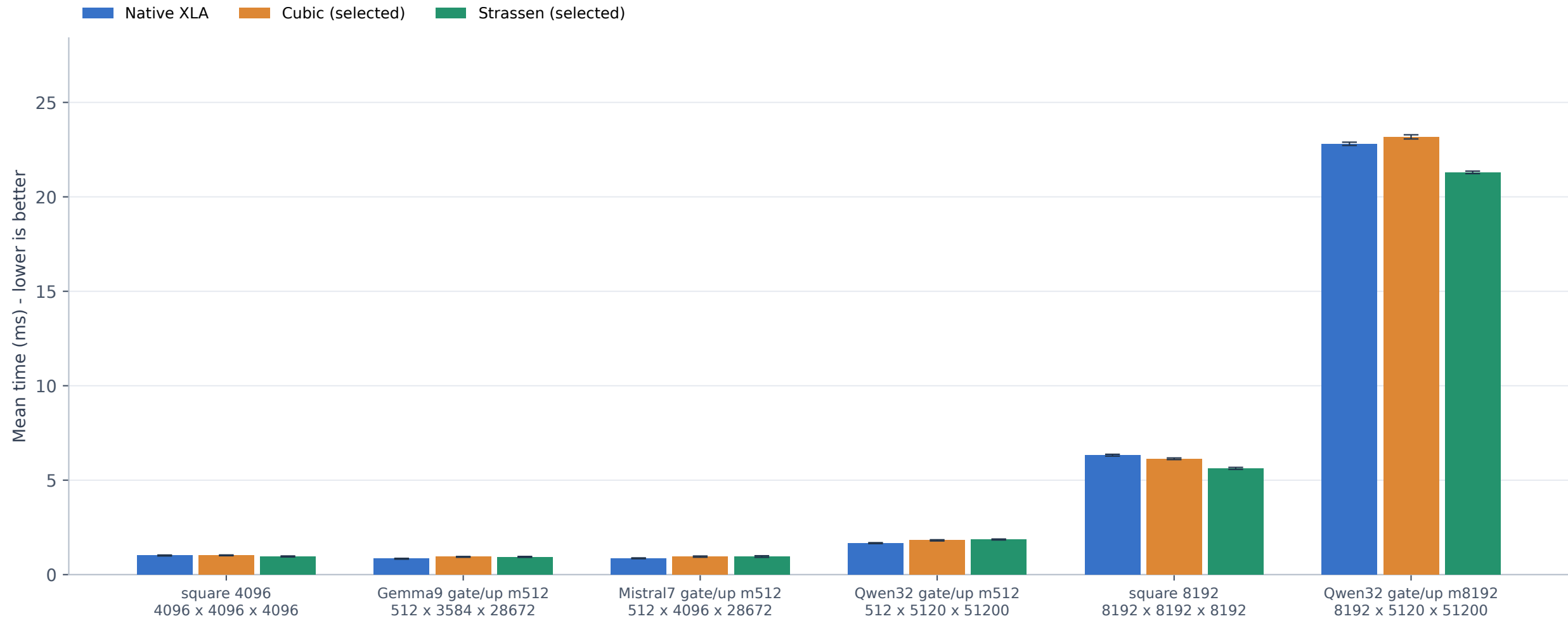


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N5 | Fresh confirmation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | N5-N9 allocation

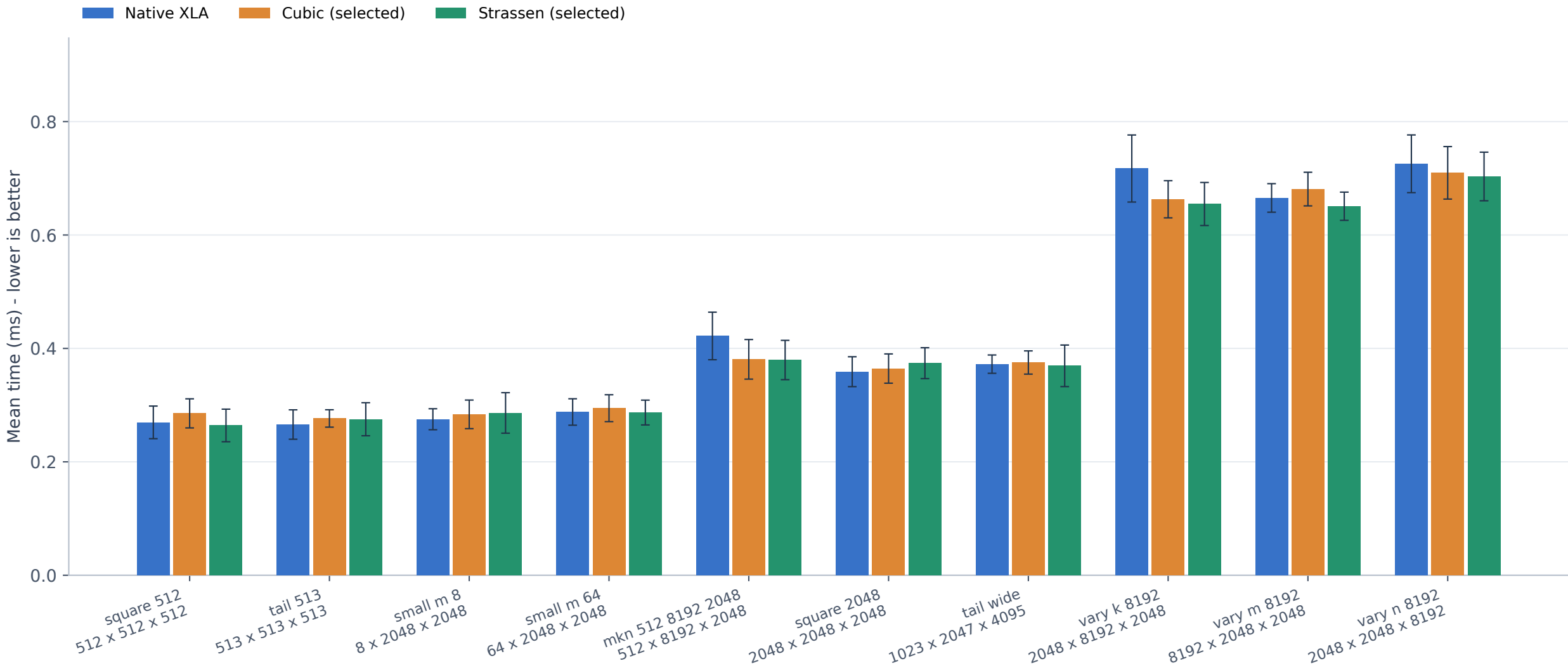


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N5 | Fresh confirmation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | N5-N9 allocation

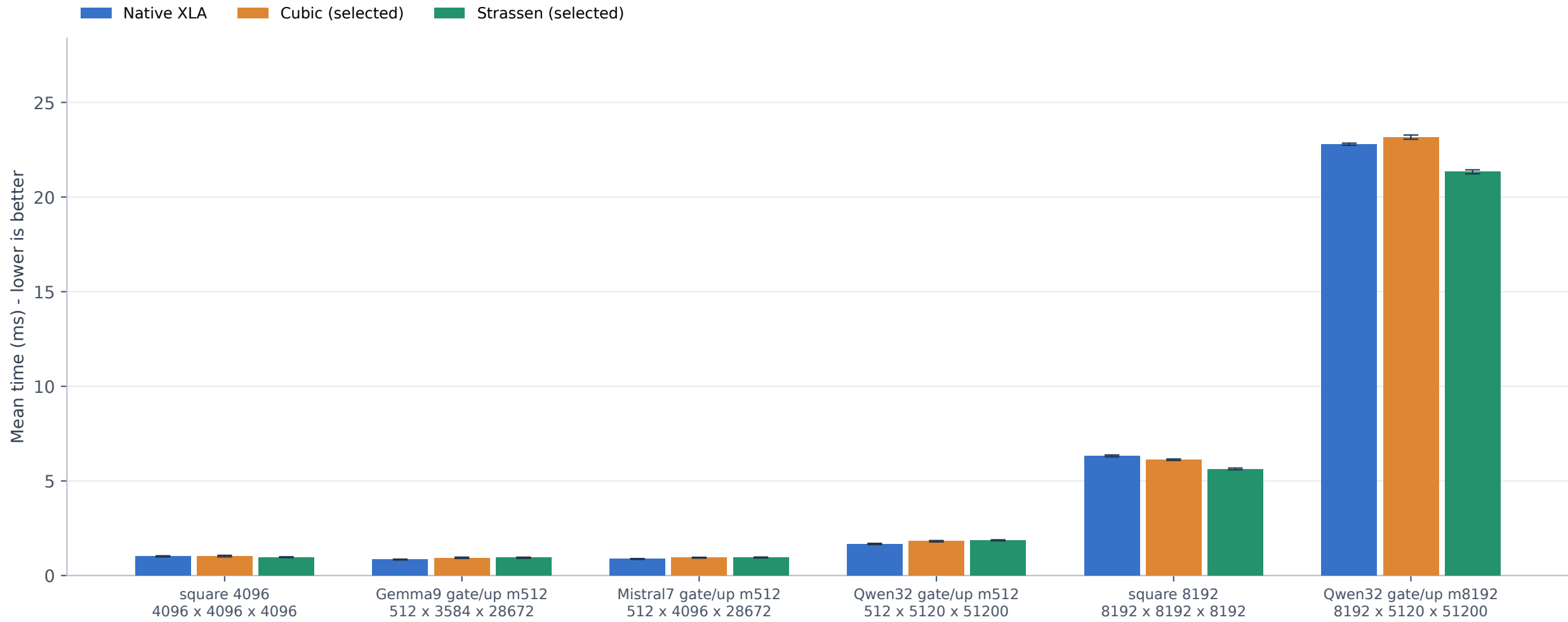


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N5 | Fresh confirmation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | N5-N9 allocation

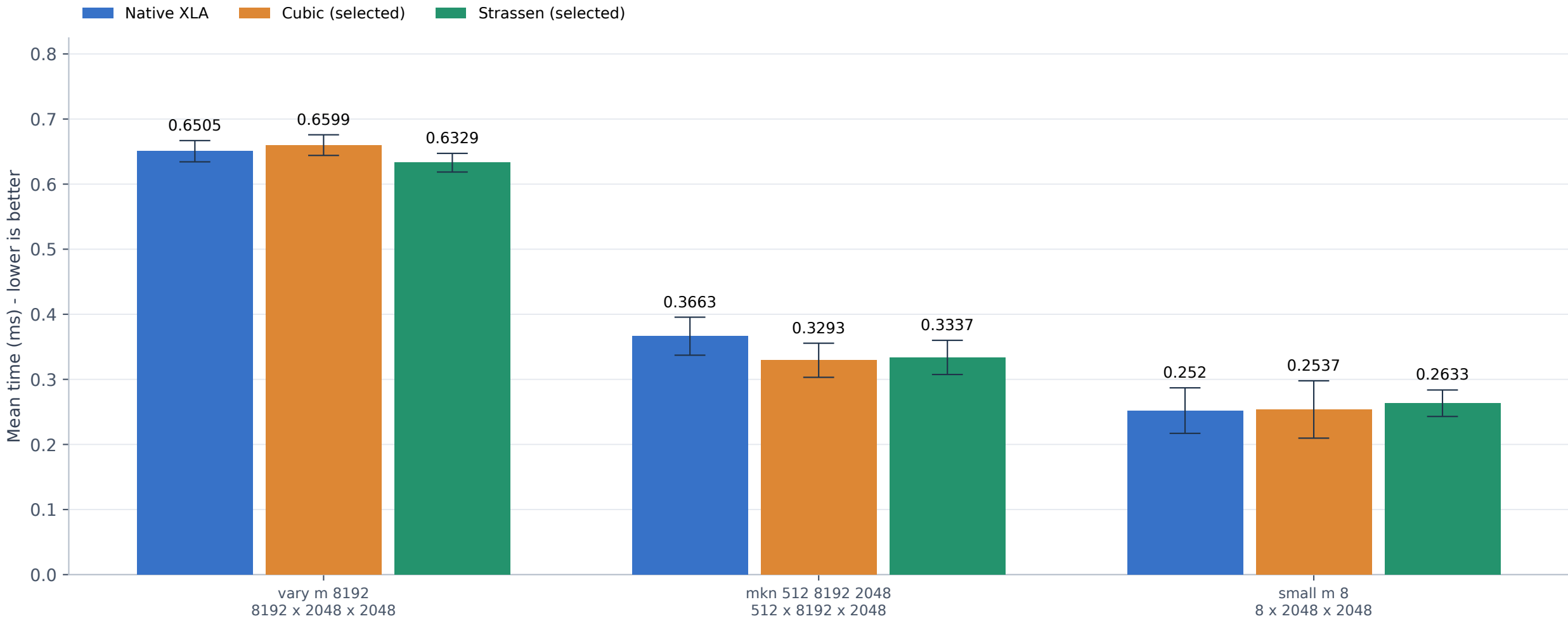


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N6 | Registered representatives - before profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.

Complete call | N5-N9 allocation

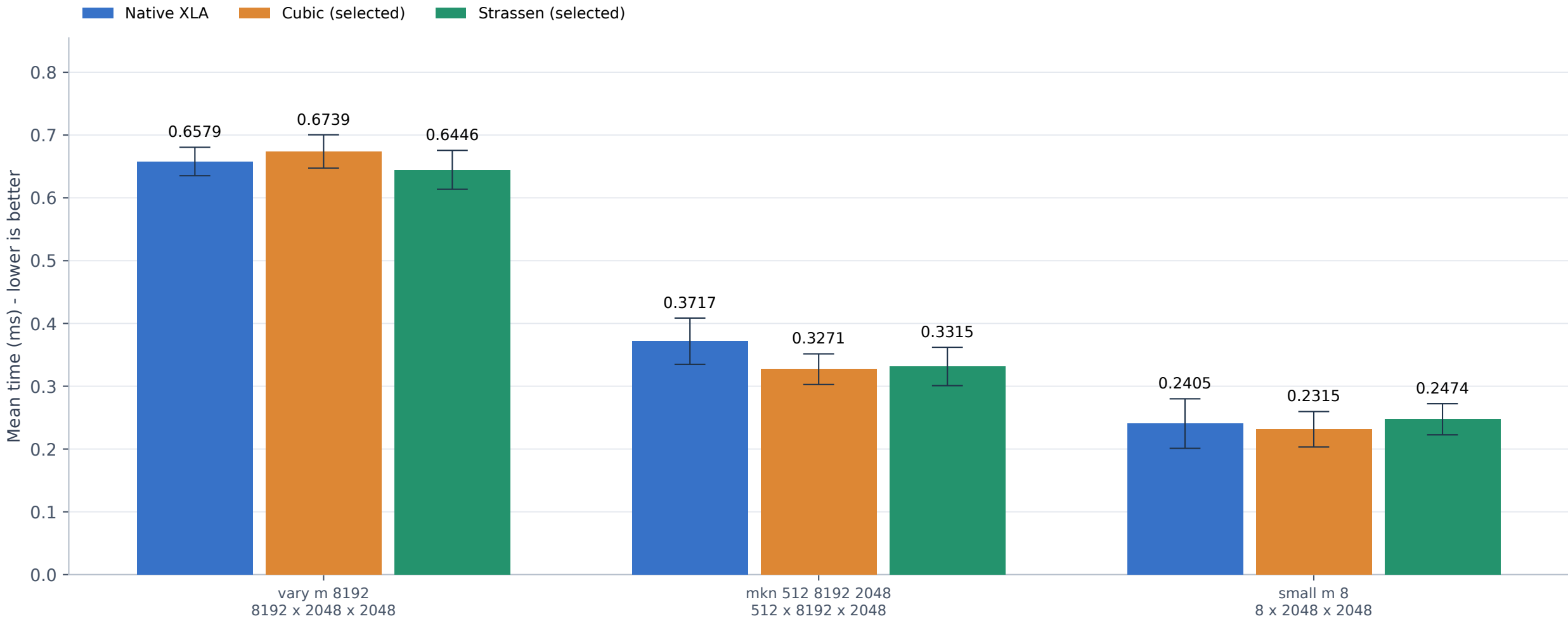


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N6 | Registered representatives - after profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.

Complete call | N5-N9 allocation

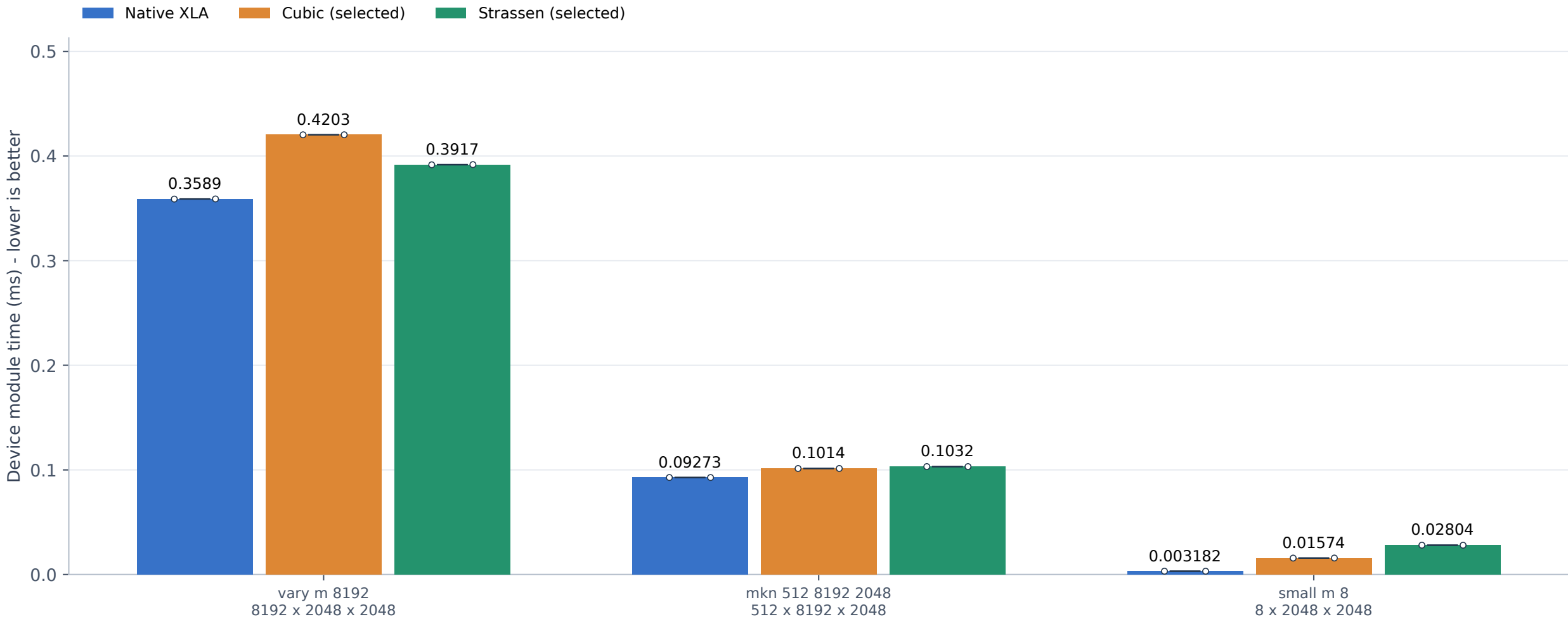


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N6 | Profiled TPU module

Mean of two 8-call captures; whiskers/dots show capture means. Profiler-instrumented, not ordinary call latency.

Profiled TPU module | N5-N9 allocation

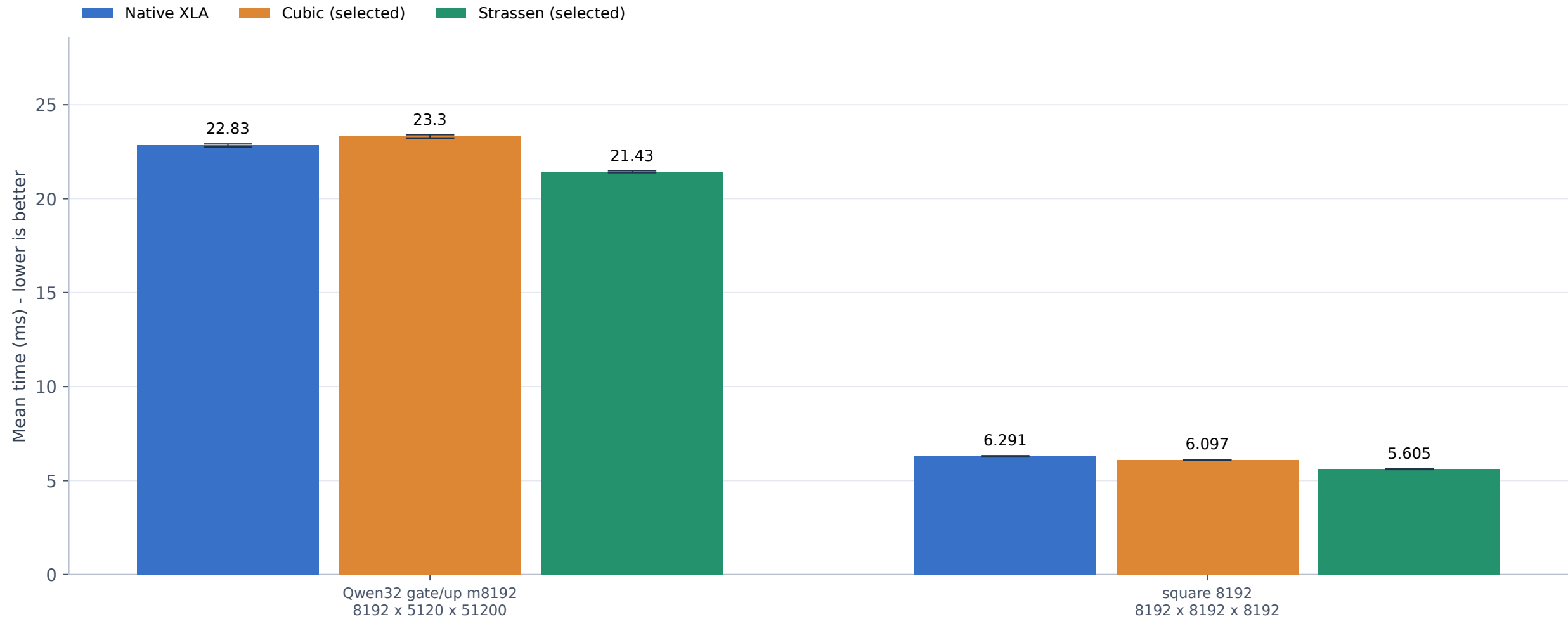


Capture-mean range is descriptive; no utilization, overlap or HBM-traffic claim.

N6 | Large-shape supplement - before profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.

Complete call | N5-N9 allocation

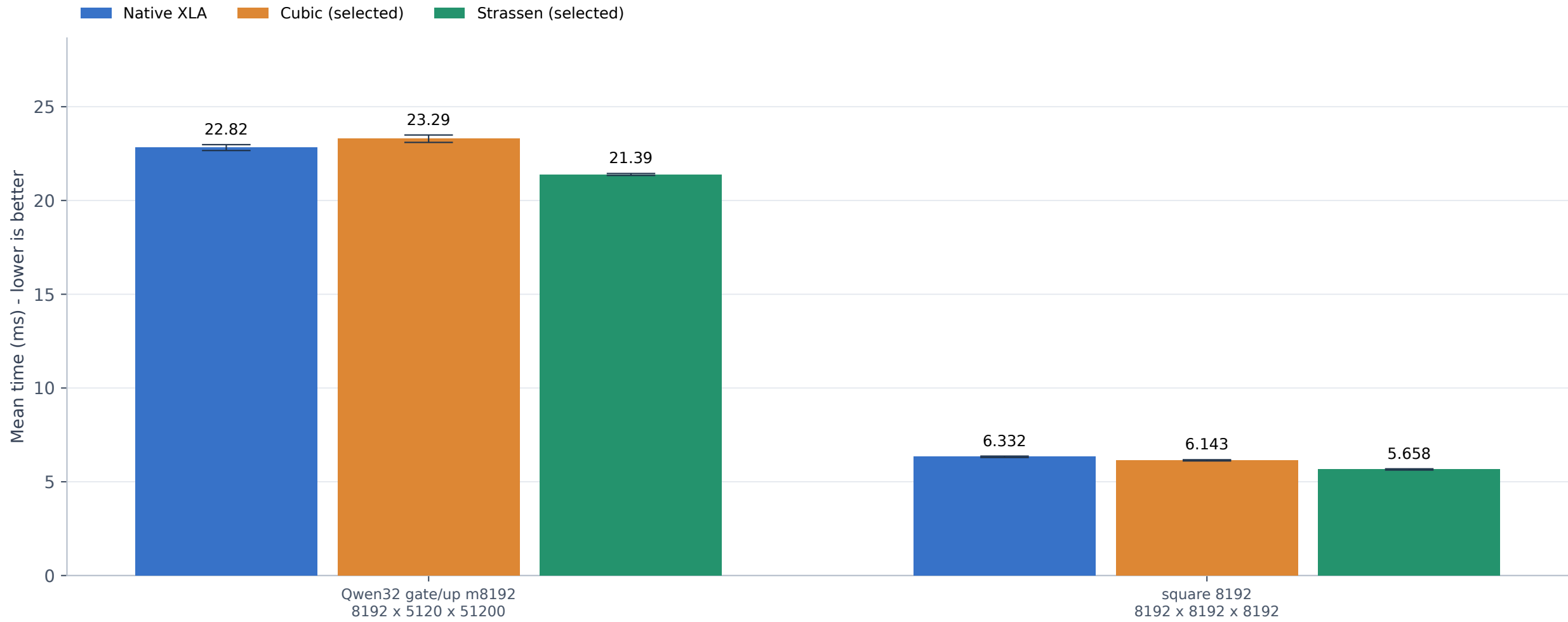


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N6 | Large-shape supplement - after profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.

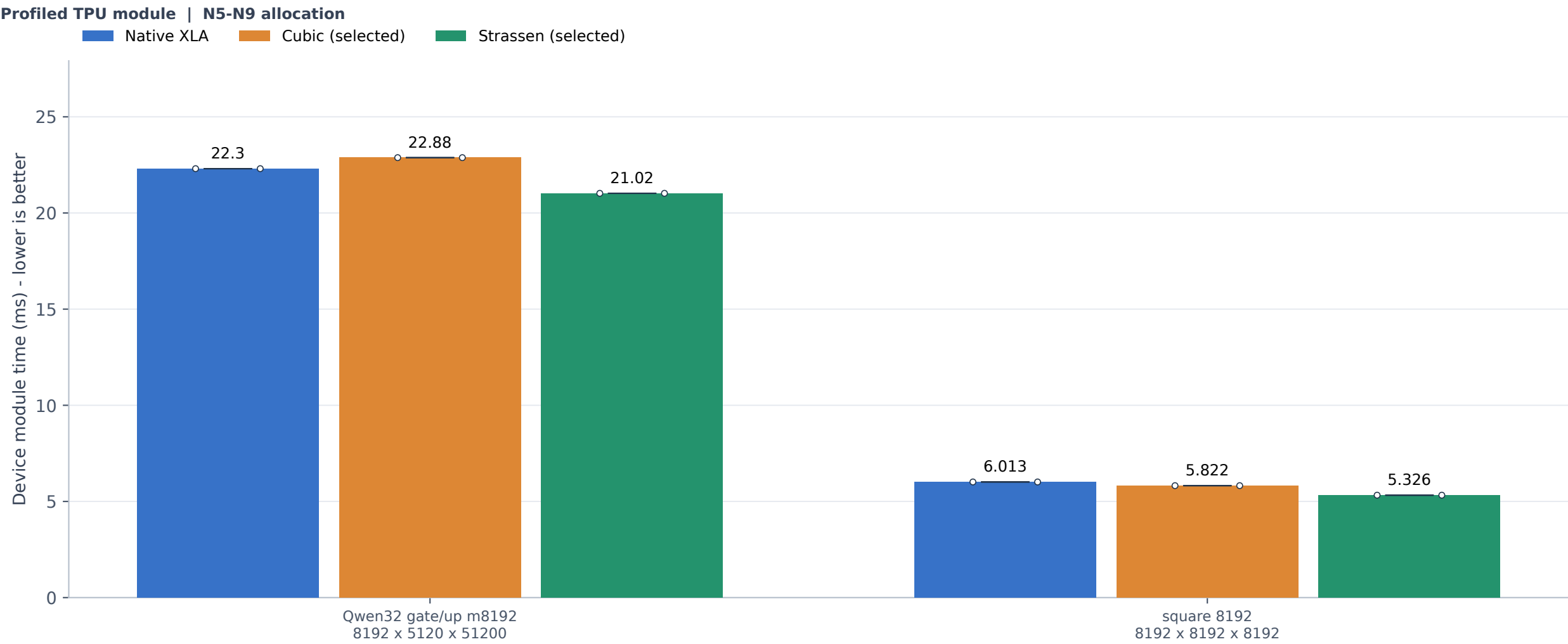
Complete call | N5-N9 allocation



Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N6 | Profiled TPU module - large-shape supplement

Mean of two 8-call captures; whiskers/dots show capture means. Profiler-instrumented, not ordinary call latency.

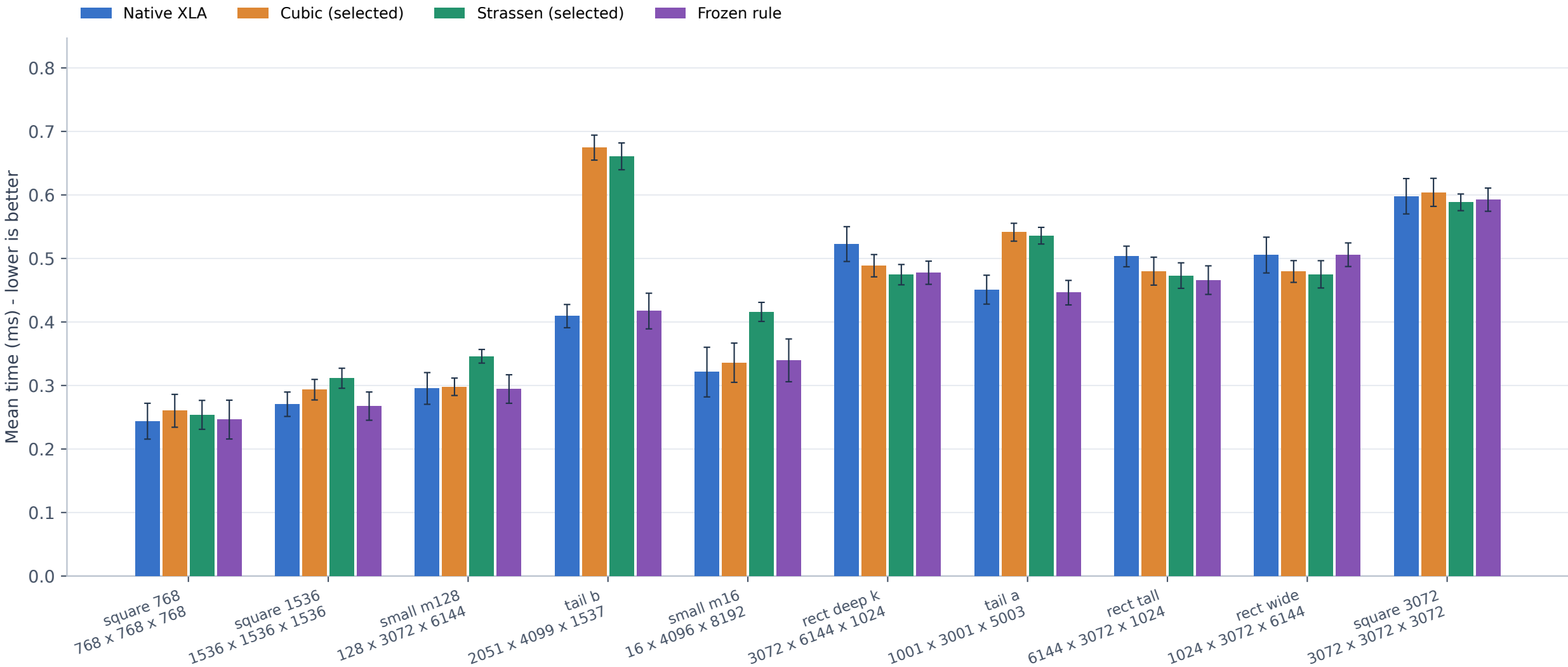


Capture-mean range is descriptive; no utilization, overlap or HBM-traffic claim.

N7 | Frozen-rule evaluation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | N5-N9 allocation

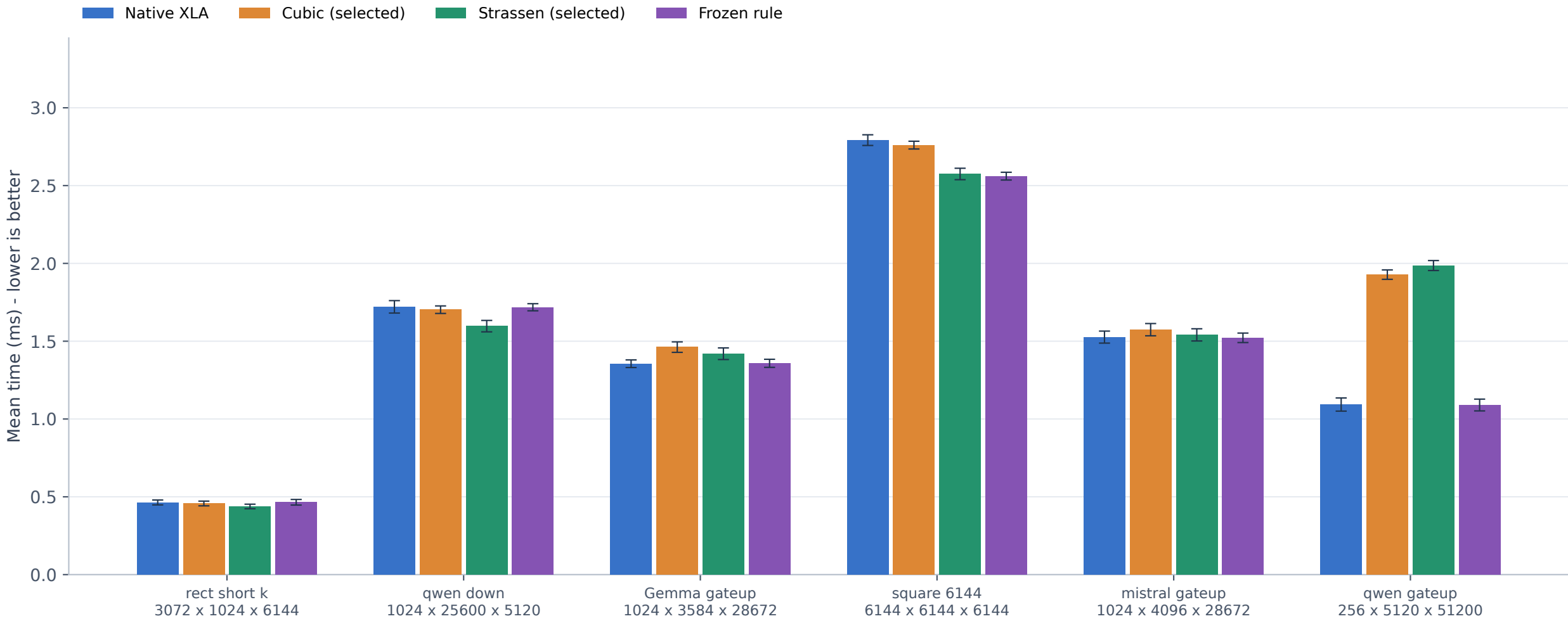


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Frozen-rule evaluation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | N5-N9 allocation

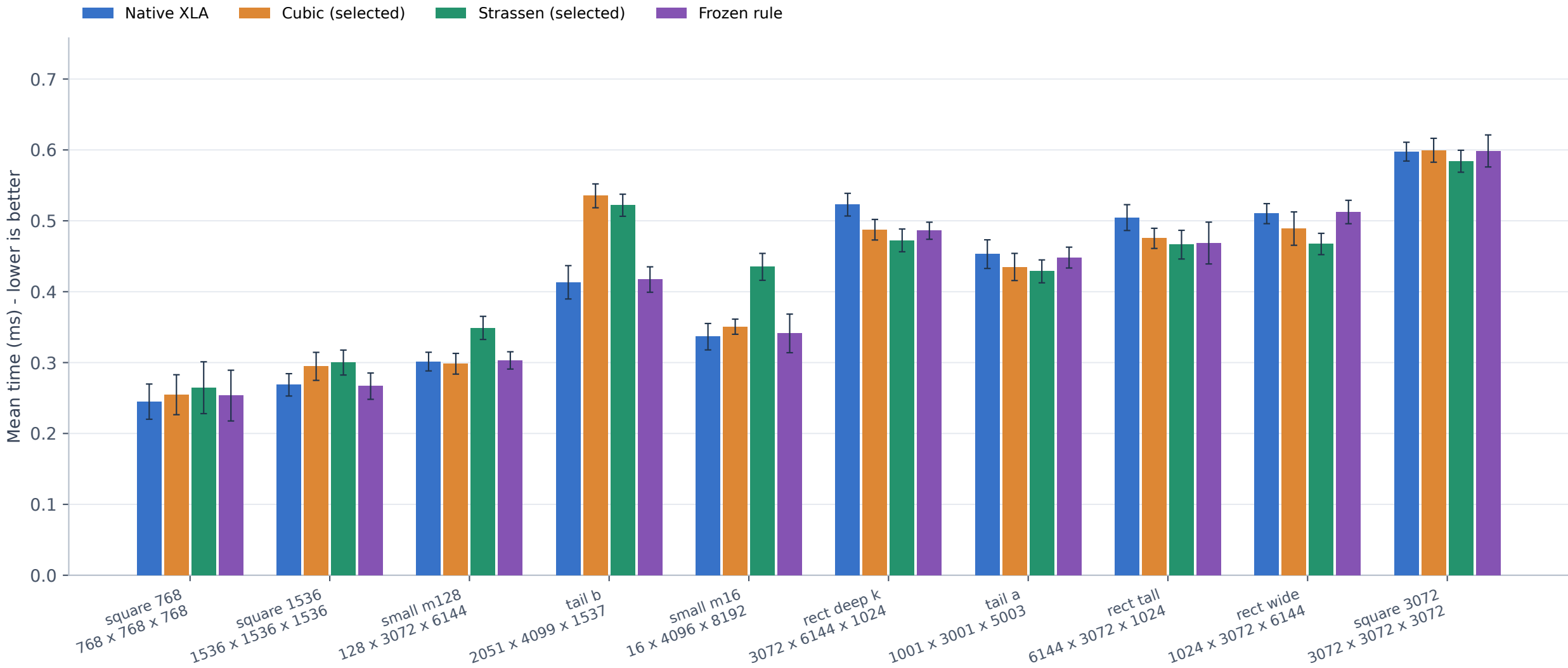


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Frozen-rule evaluation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | N5-N9 allocation

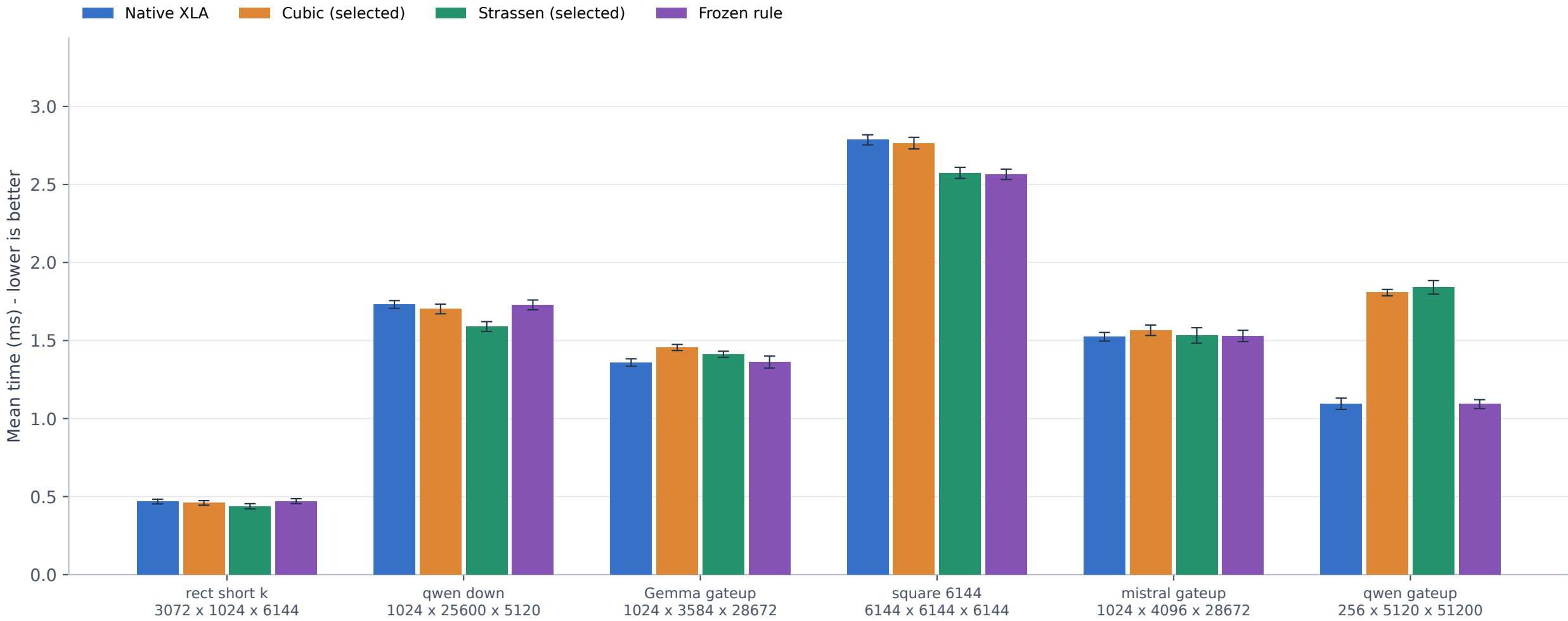


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Frozen-rule evaluation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | N5-N9 allocation

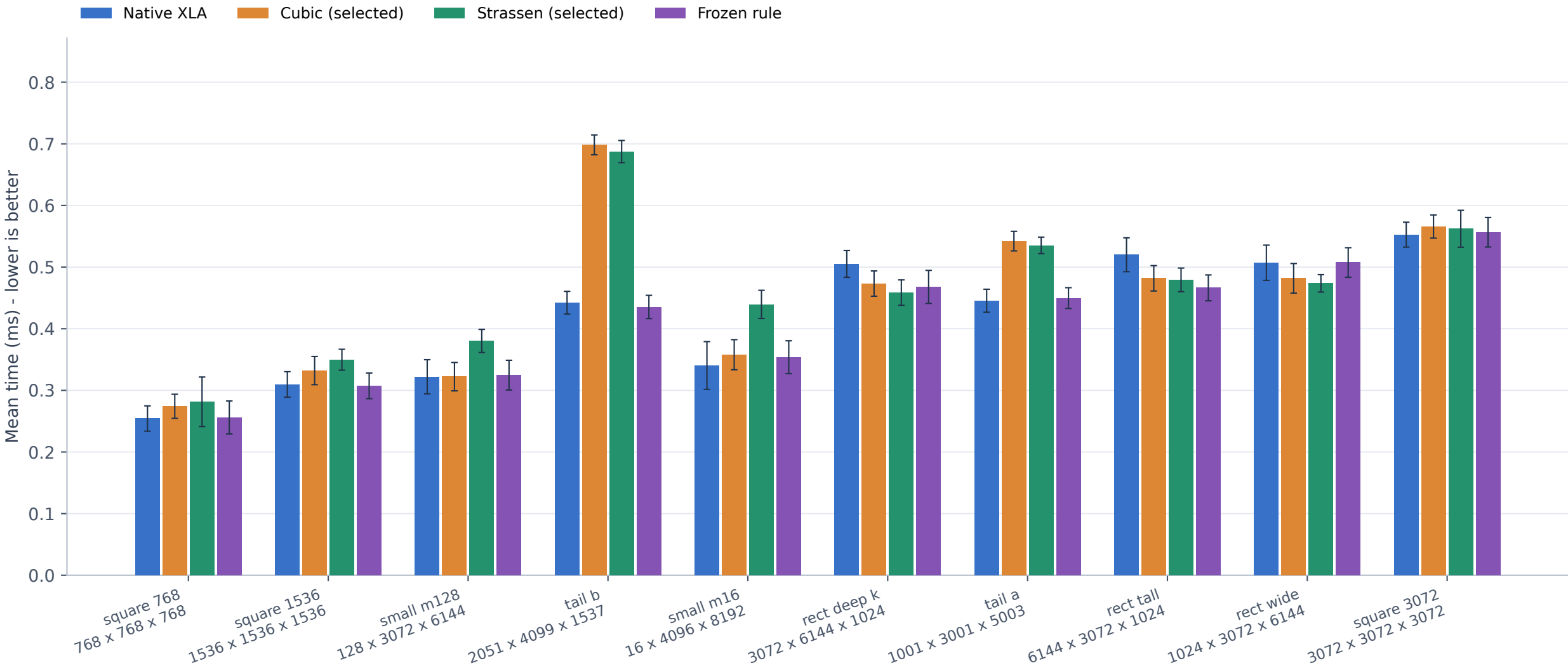


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Fresh-v5e replication - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | Fresh v5e replication

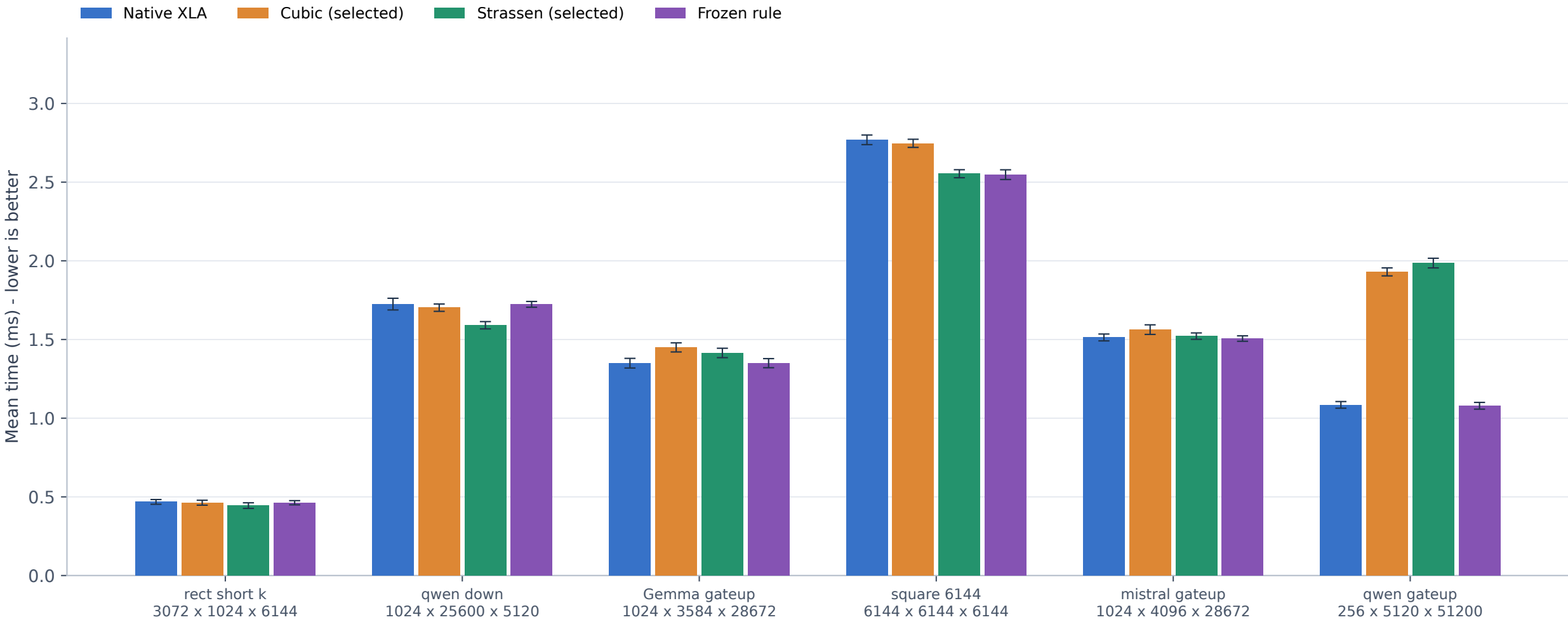


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Fresh-v5e replication - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | Fresh v5e replication

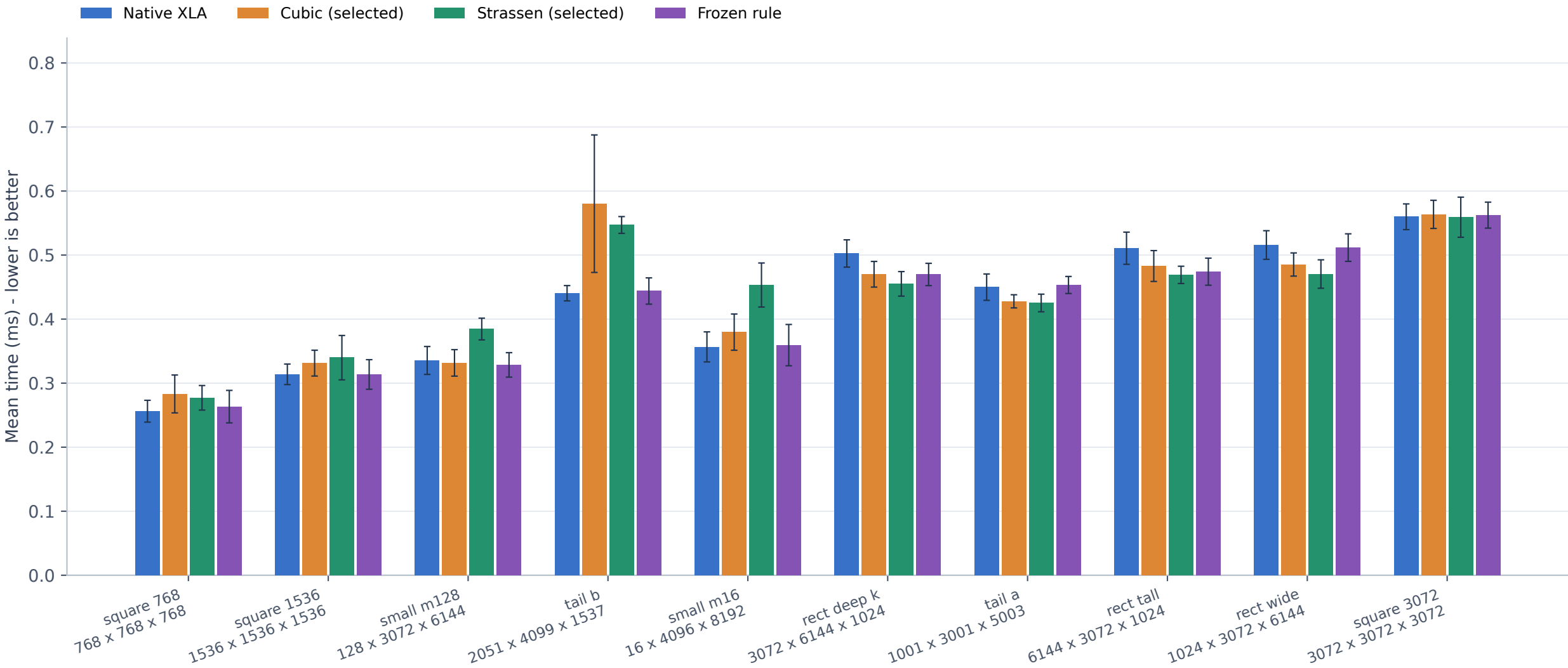


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Fresh-v5e replication - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | Fresh v5e replication

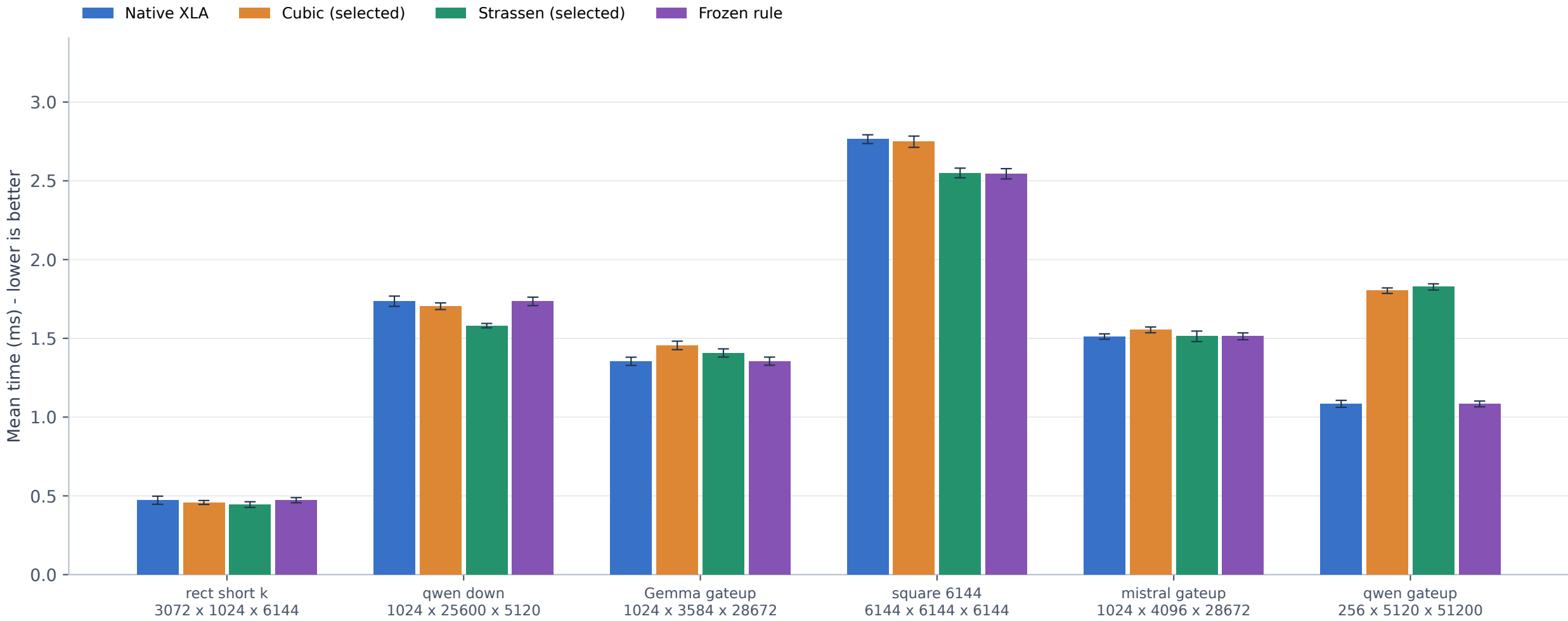


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N7 | Fresh-v5e replication - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | Fresh v5e replication

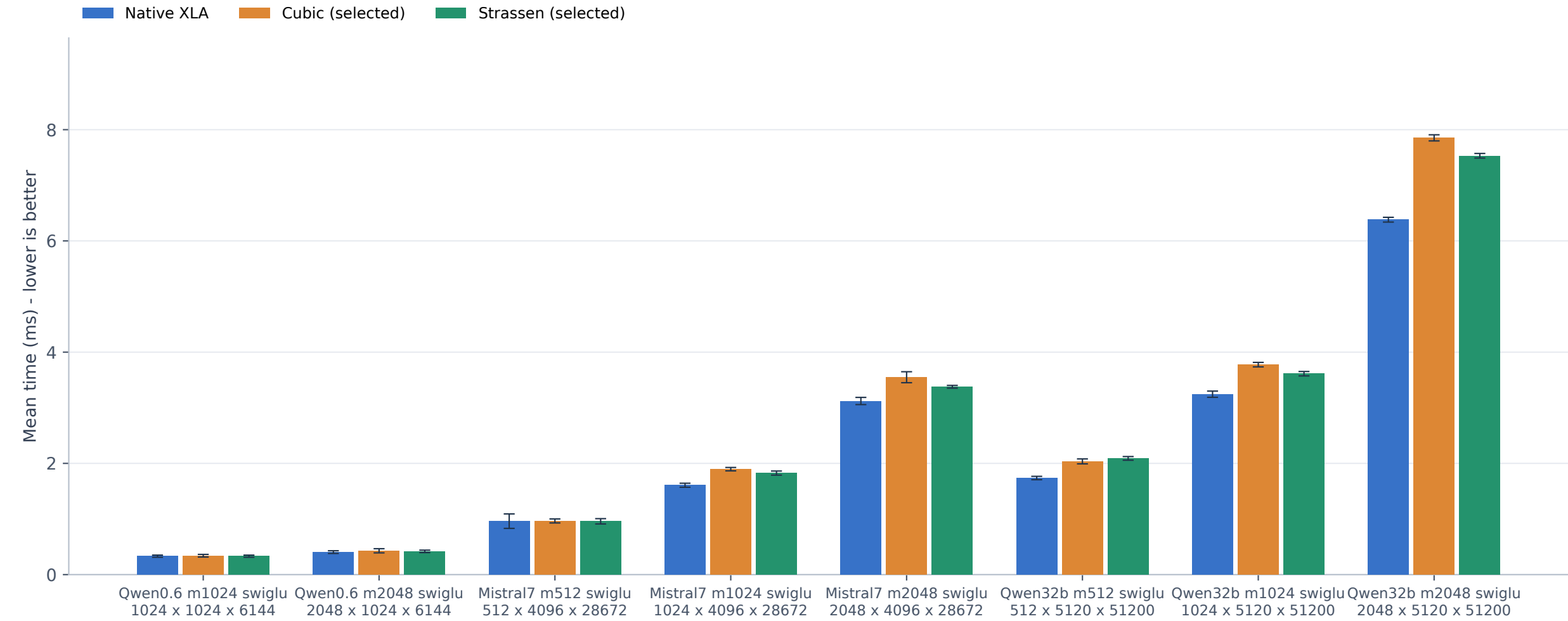


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N8 | swiglu - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.

Complete call | N5-N9 allocation

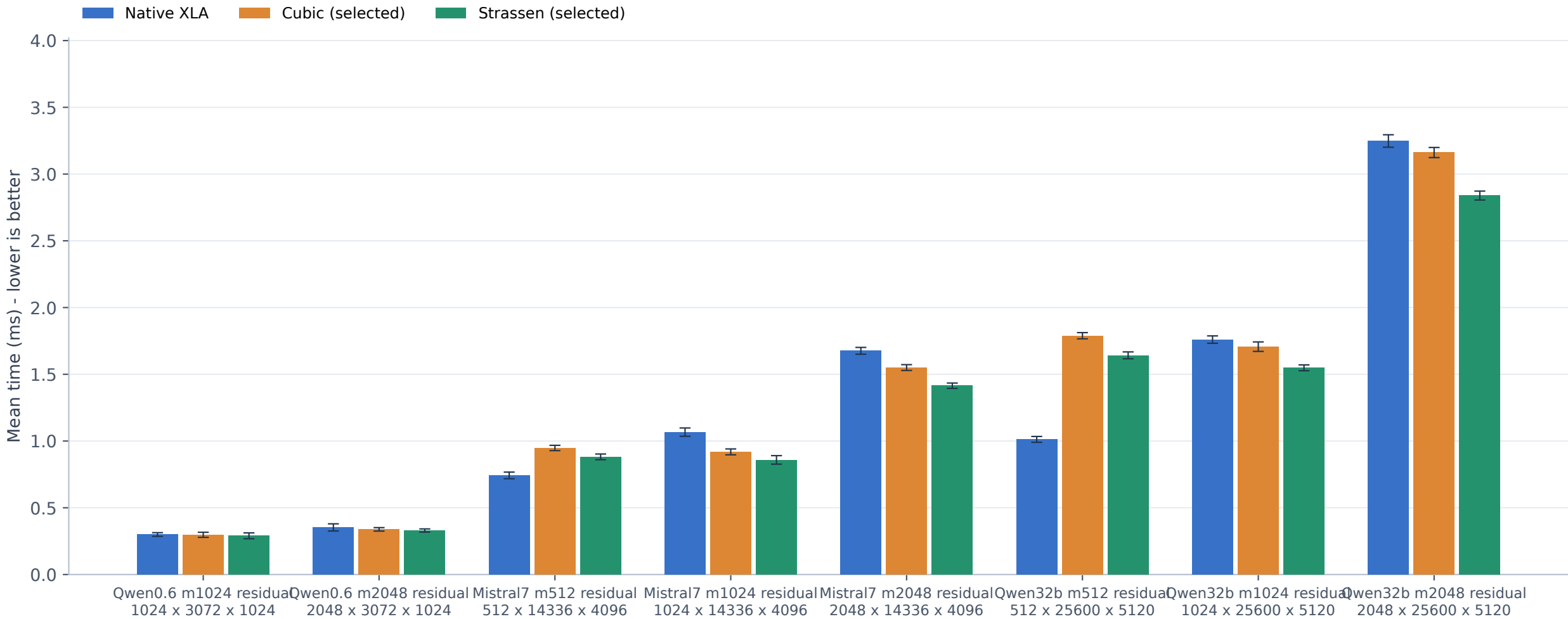


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N8 | residual - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.

Complete call | N5-N9 allocation

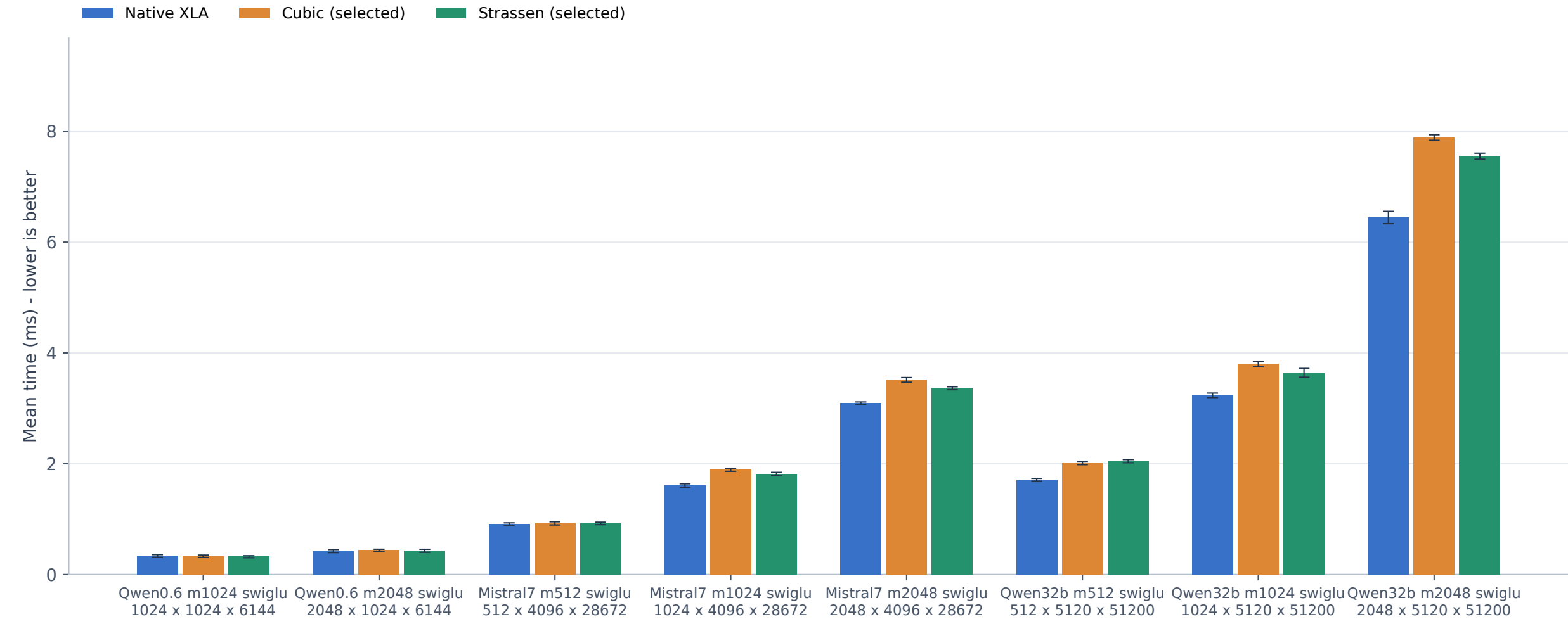


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N8 | swiglu - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.

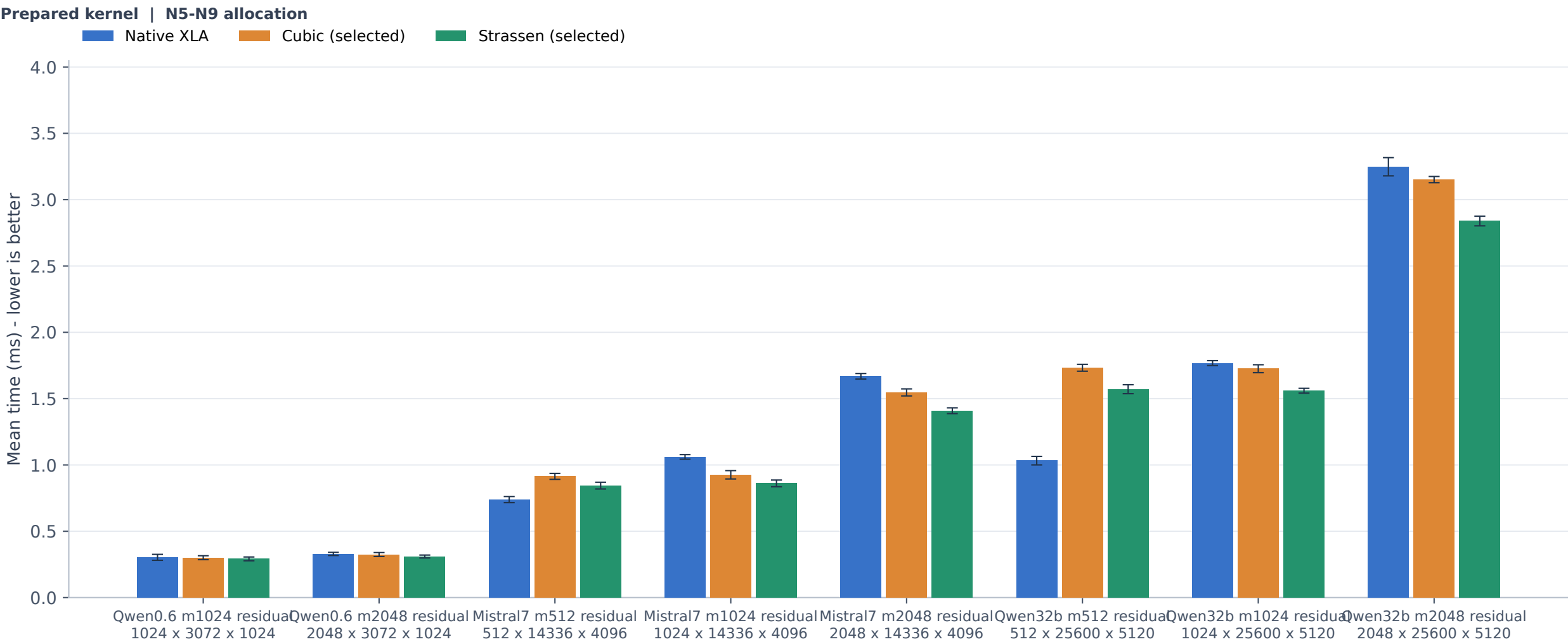
Prepared kernel | N5-N9 allocation



Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N8 | residual - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.

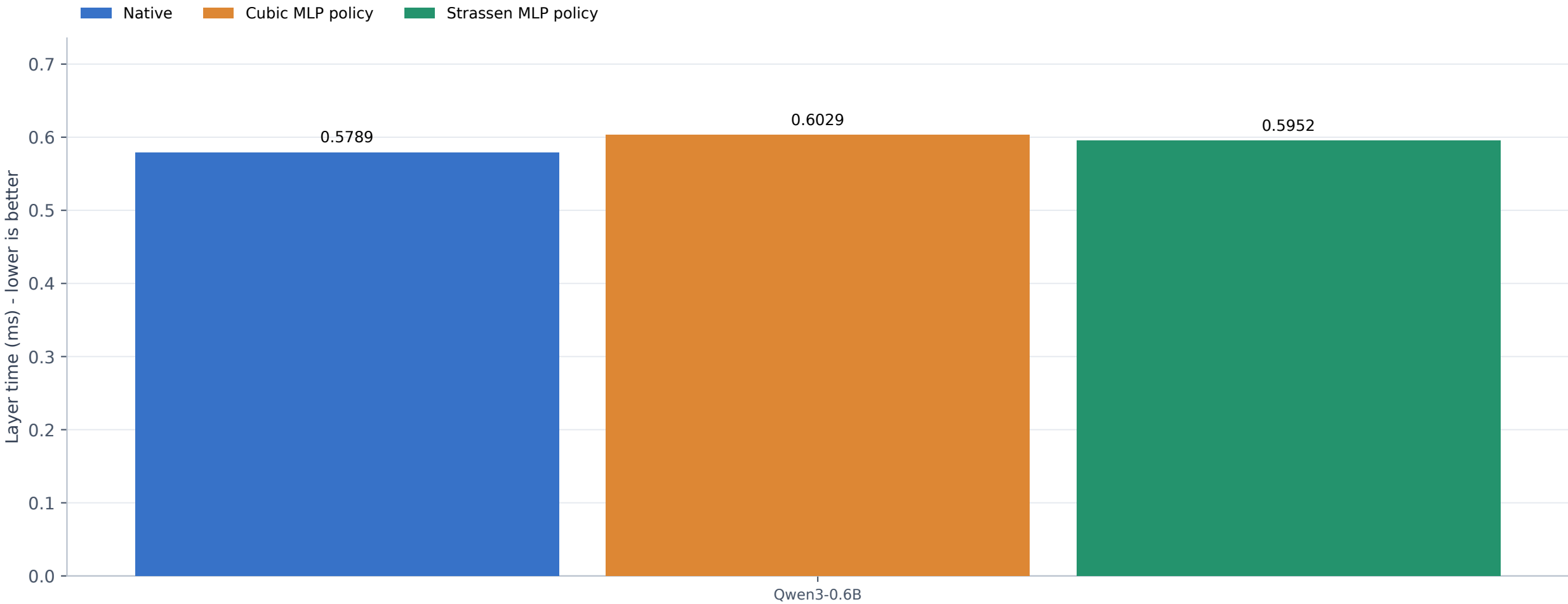


Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

N9 | Qwen3-0.6B - resident layer

Complete transformer layer; weights resident. Mean across 28 or 32 different layers; layers are not independent replicates.

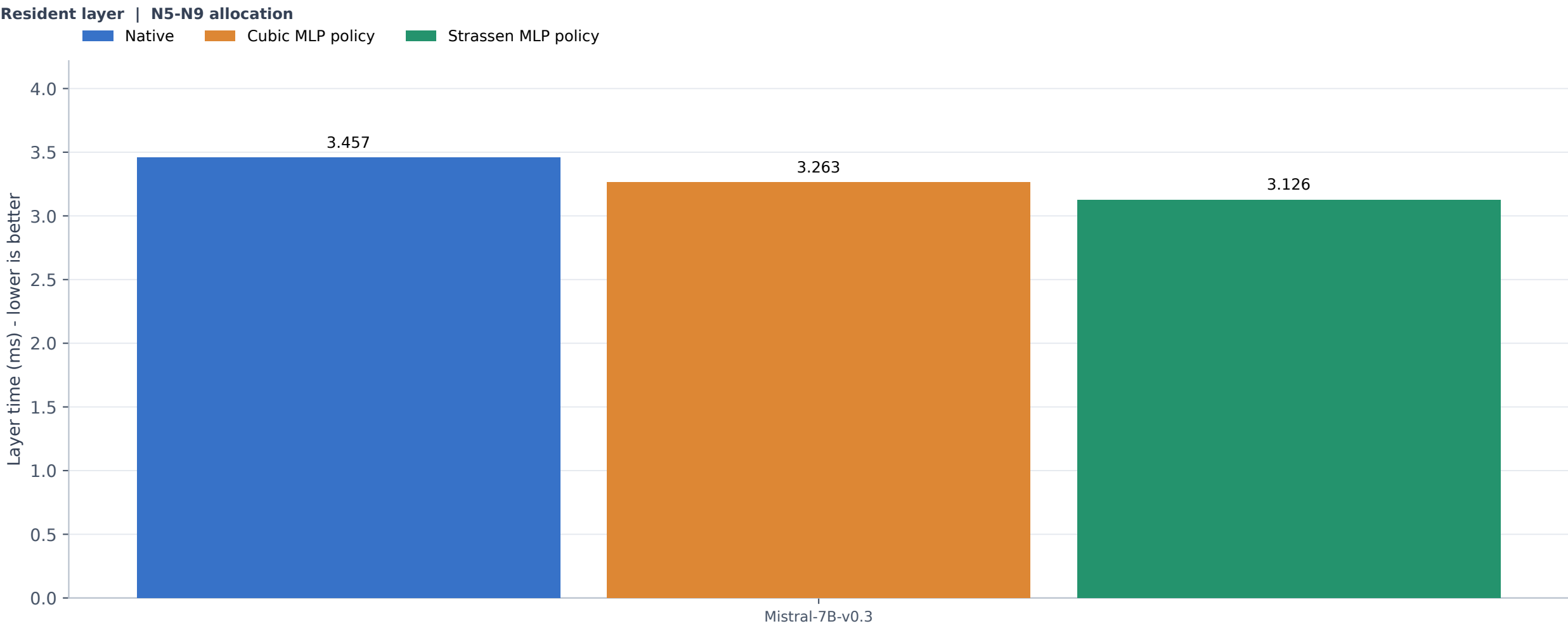
Resident layer | N5-N9 allocation



No pooled layer confidence interval is inferred.

N9 | Mistral-7B-v0.3 - resident layer

Complete transformer layer; weights resident. Mean across 28 or 32 different layers; layers are not independent replicates.

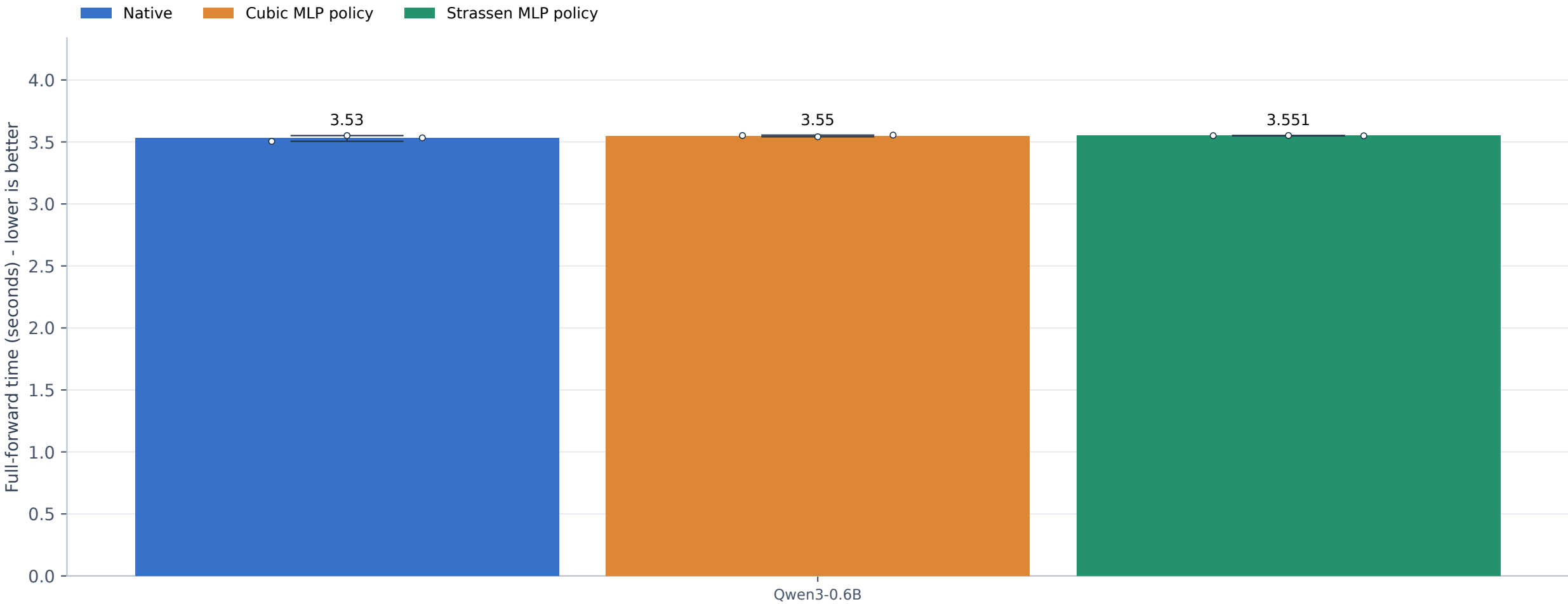


No pooled layer confidence interval is inferred.

N9 | Qwen3-0.6B - streamed full forward

1,024-token full forward including reads, host layout and transfers. All three measured repeats are shown.

Streamed model | N5-N9 allocation

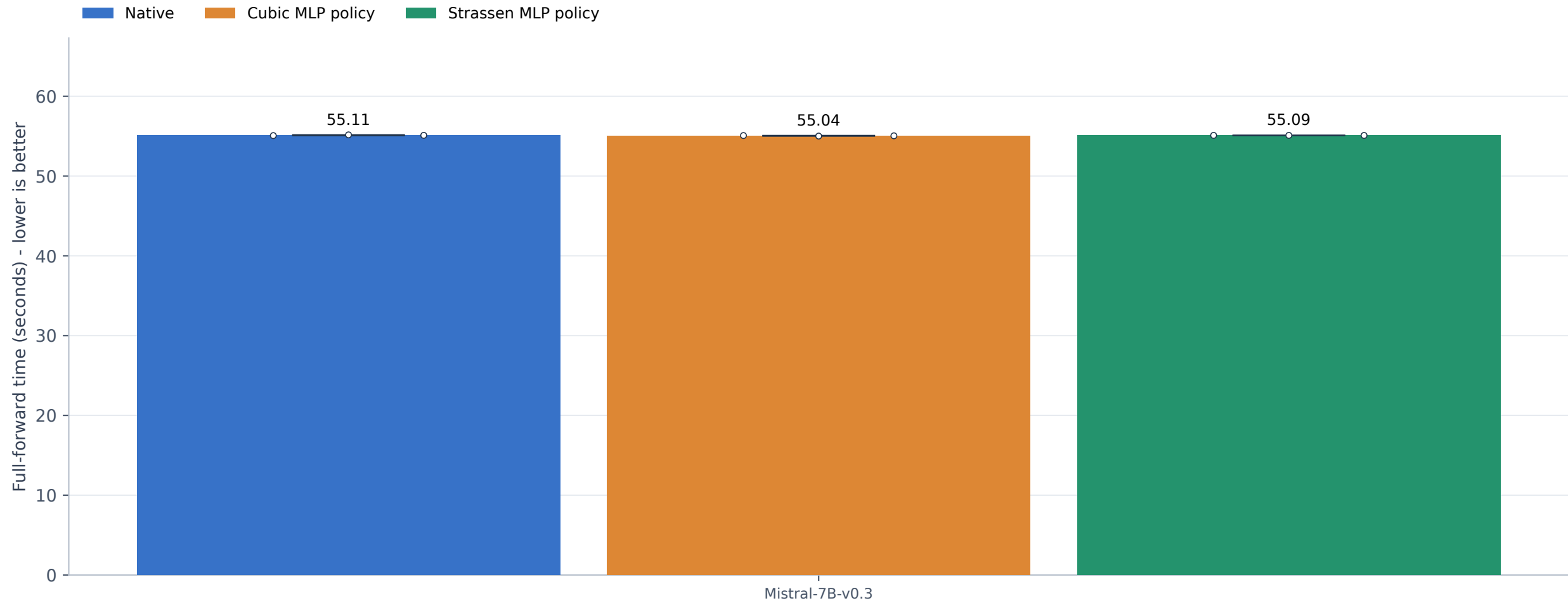


Whiskers: minimum to maximum of 3 repeats; descriptive, not a confidence interval.

N9 | Mistral-7B-v0.3 - streamed full forward

1,024-token full forward including reads, host layout and transfers. All three measured repeats are shown.

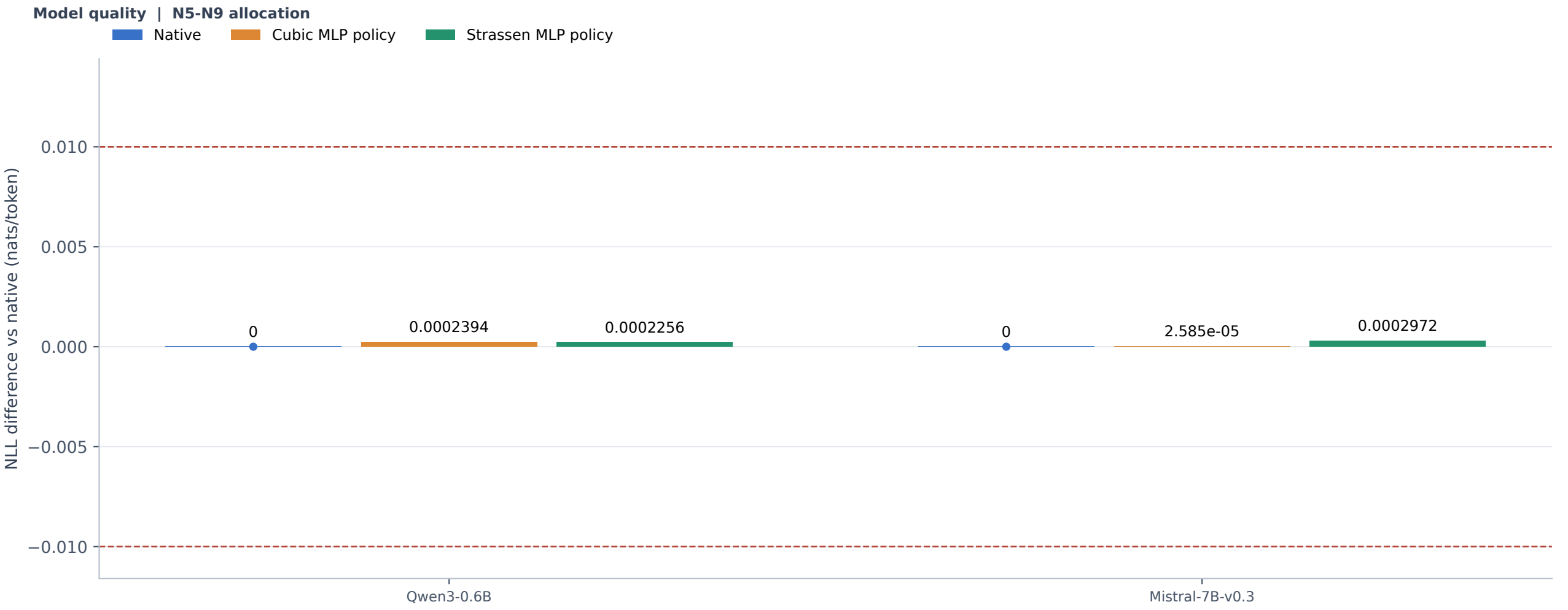
Streamed model | N5-N9 allocation



Whiskers: minimum to maximum of 3 repeats; descriptive, not a confidence interval.

N9 | Corpus quality - nll delta

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.



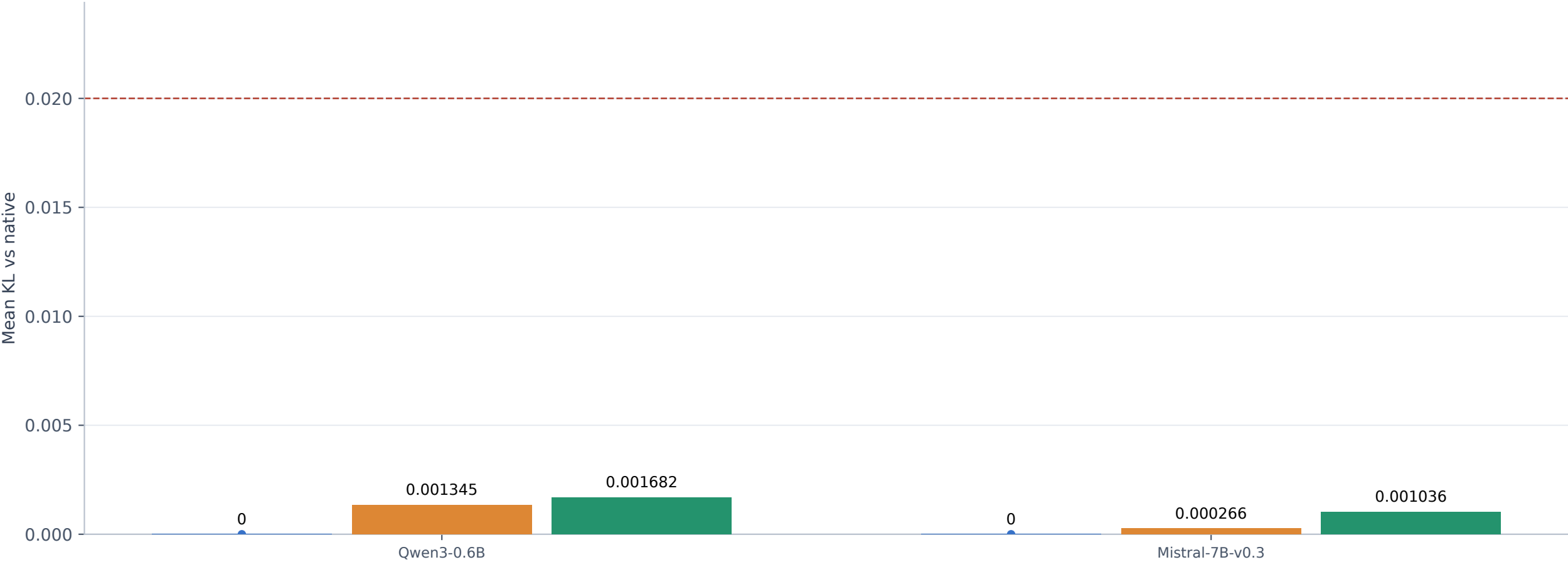
Dotted line: predeclared aggregate gate. No statistical interval is inferred.

N9 | Corpus quality - mean kl

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.

Model quality | N5-N9 allocation

Native Cubic MLP policy Strassen MLP policy

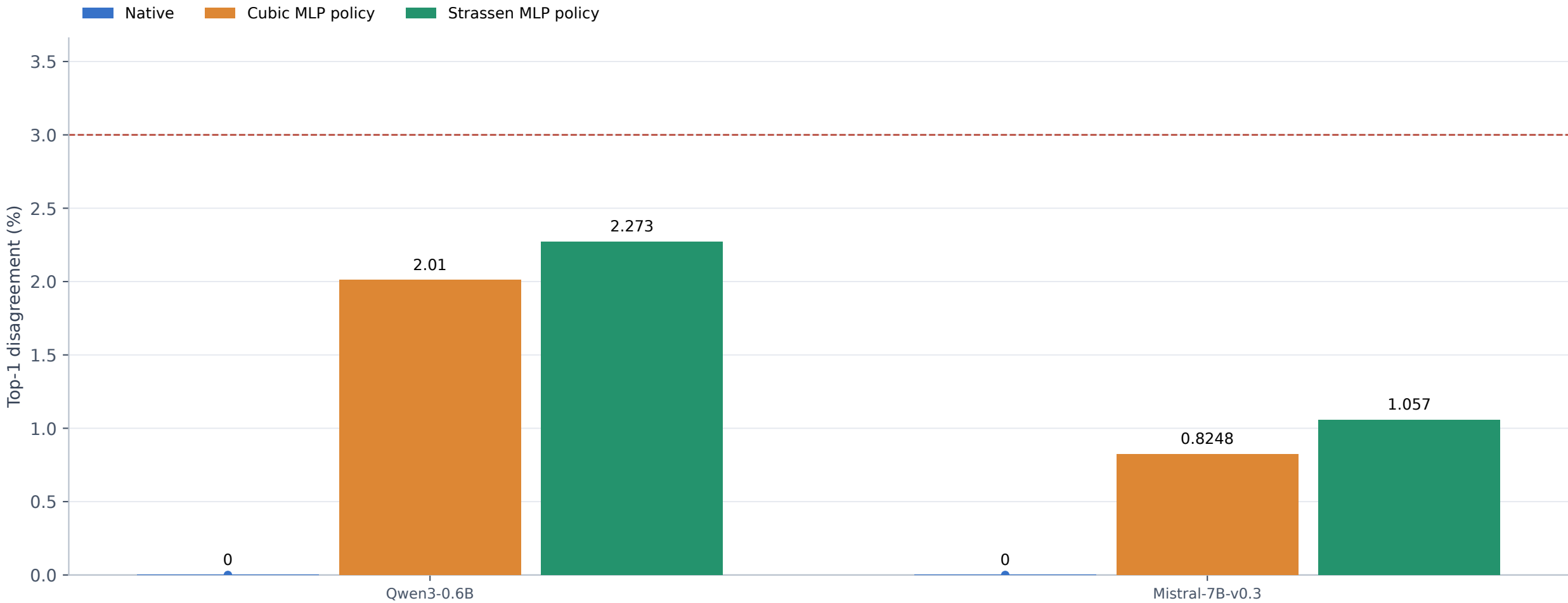


Dotted line: predeclared aggregate gate. No statistical interval is inferred.

N9 | Corpus quality - top1 disagreement

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.

Model quality | N5-N9 allocation



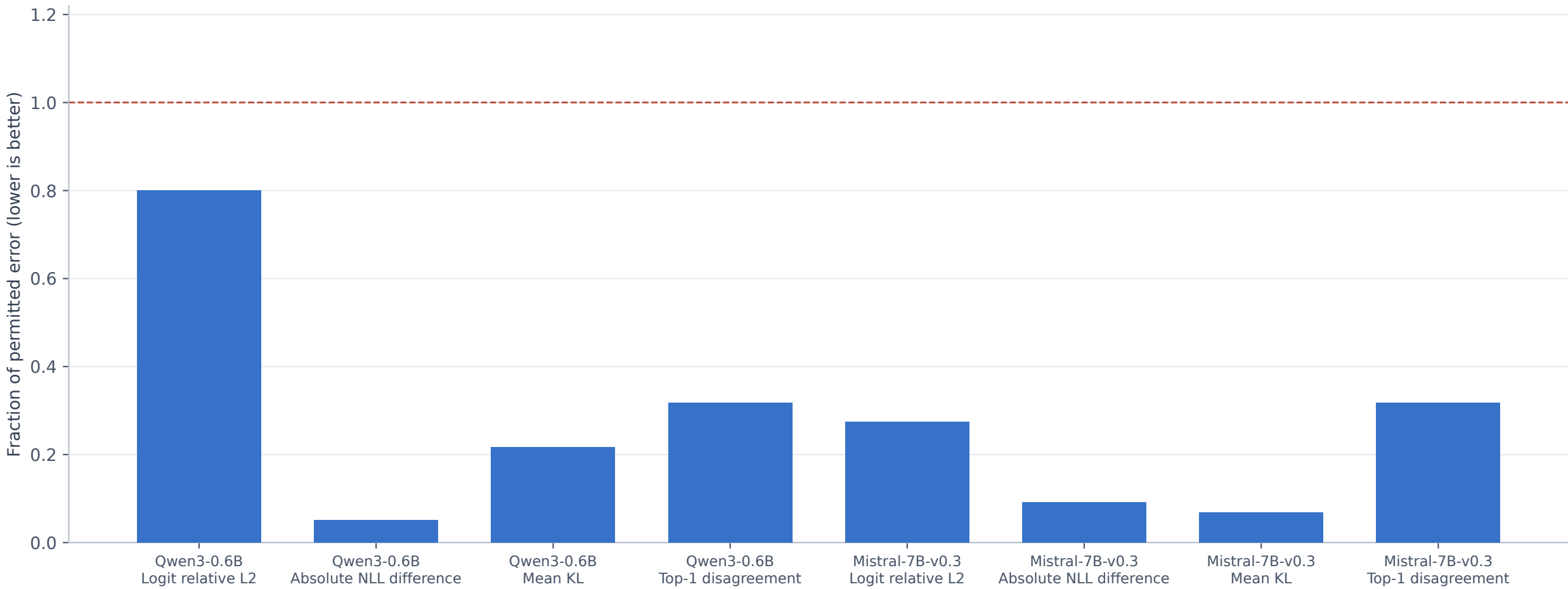
Dotted line: predeclared aggregate gate. No statistical interval is inferred.

N9 | Independent native implementation qualification

Native JAX vs official Transformers CPU BF16 on 64 input tokens / 63 scored positions.

Model quality | N5-N9 allocation

Native JAX / allowed error



The limits differ from the full-corpus custom-policy gates. Finite-output checks also passed.